INTERNAL IMPLEMENTATION SPECIFICATION

Alec's x ScaleSight

Alec's Retail-to-Production Planning Workspace

Product and Technical Specification

Developer Implementation Guide for Wahib

Version 1.0 | Prepared for ScaleSight
Specification baseline: 12 October 2026
Evidence research snapshot: 11 October 2026

Retail signals, reviewed against production, components and cash.

Analysis maintained by ScaleSight.
Decisions made with Alec's.

Public context, synthetic operating inputs, derived metrics and analyst interpretation remain distinct. No live connection or client results.

Contents

01 Document control & reading guide 4

02 Objective, positioning & founder walkthrough 5

03 Public context & evidence boundaries 6

04.01 Phase 1 / Canonical timeline and units 9

04.02 Phase 1 / Canonical public product catalog 10

04.03 Phase 1 / Operating scope and retailer eligibility 12

04.04 Phase 1 / SKU × retailer demand dataset 13

04.05 Phase 1 / Historical SKU dataset 14

04.06 Phase 1 / Store-cohort dataset 17

04.07 Phase 1 / Counterfactual dataset 20

04.08 Phase 1 / Launch fill, reorders and retailer inventory 22

04.09 Phase 1 / Forward demand and dispatch assumptions 24

04.10 Phase 1 / Canonical finished-goods position 27

04.11 Phase 1 / Production commitments 28

04.12 Phase 1 / Later production receipts 30

04.13 Phase 1 / Reconciled inventory trajectory 31

04.14 Phase 1 / Planning BOM 34

04.15 Phase 1 / Component availability dataset 36

04.16 Phase 1 / Component purchase orders 38

04.17 Phase 1 / Retailer × SKU allocation dataset 41

04.18 Phase 1 / SKU economics and cash 43

04.19 Phase 1 / Forecast-learning dataset 46

04.20 Phase 1 / Scenario dataset 48

04.21 Phase 1 / Calculation engine 51

04.22 Phase 1 / Recommendation logic and confidence 53

05 Data dictionary, relationships & engine contracts 54

06.01 Data brief / Weekly Decision Brief 58

06.02 Data brief / Launch Incrementality 60

06.03 Data brief / Reorder Quality 61

06.04 Data brief / Store Cohorts 62

06.05 Data brief / Production Priorities 63

06.06 Data brief / Components & Commitments 64

06.07 Data brief / Allocation 65

06.08 Data brief / Cash Exposure 66

06.09 Data brief / Forecast Learning 67

06.10 Data brief / Scenario Lab 68

06.11 Data brief / Managed Planning 69

07.01 Phase 2 / Brand direction 71

07.02 Phase 2 / Navigation & shared state 73

07.03 Page 1 / Weekly Decision Brief 77

07.04 Page 2 / Launch Incrementality 80

07.05 Page 3 / Reorder Quality 83

07.06 Page 4 / Store Cohorts 86

07.07 Page 5 / Production Priorities 88

07.08 Page 6 / Components & Commitments 91

07.09 Page 7 / Allocation 94

07.10 Page 8 / Cash Exposure 96

07.11 Page 9 / Forecast Learning 98

07.12 Page 10 / Scenario Lab 101

07.13 Page 11 / Managed Planning 104

08 Shared presentation, responsive & blocked states 107

09 Build sequence & technical acceptance 111

10.A Source register 115

10.B Assumptions & unresolved implementation points 116

10.C Deterministic store-panel generation specification 118

10.D Formula and selector reference 120

01 Document control & reading guide

Prepared for: ScaleSight **Intended reader:** Wahib, developer **Version:** 1.0 - implementation baseline
Specification baseline: 12 October 2026 **Evidence research snapshot:** 11 October 2026

> Illustrative planning concept. Operating data and recommendations are synthetic. No live connection or client results.

The document contains all eleven routes. A four-to-six-minute walkthrough is a navigation path, not a smaller build scope. The Fishwife specification supplies the document architecture and engineering-depth reference; none of its operating data is reused.

Source precedence: explicit executable instructions; complete Phase 1 for numbers and logic; latest Phase 2 for route labels and exact copy; expanded Phase 2 for additional implementation behavior. Public context is reproduced from the supplied dated research, not newly verified.

Read first: commercial objective and source context; then the complete Phase 1 tables, engine and data contracts; then the eleven data briefs and consolidated UX chapters. The source register, assumption register and formula appendix complete the implementation baseline.

Release boundary: document arithmetic checks pass. A website build must still implement and validate fixture generation, responsive layouts, interactions and all route acceptance rules. No 44,800-row store panel or demo website has been generated as part of this document.

02 Objective, positioning & founder walkthrough

The opening question remains:

> **The collaboration is selling. Should Alec’s increase total production - or change which flavors receive the existing production capacity?**

The wider recurring question is:

> **Which retailer × flavor signals deserve a change in production, components, allocation or cash commitment?**

The demo should connect five layers:

Retail evidence → analytical interpretation → supply feasibility → commercial consequences → reviewed recommendation.

It must demonstrate the full ScaleSight proposition:

• Maintained forecasting and analytical work.

• Data-science investigation of conflicting signals.

• Supply-chain review of feasible responses.

• Quantities, timing and cash implications.

• A recurring decision brief prepared alongside the client’s team.

Alec’s already manufactures and manages a growing business. We are demonstrating additional analytical capacity around that operation - not claiming that Alec’s cannot plan.

Account classification: Type 2 working hypothesis. Existing production capability is visible; a specific internal capacity gap remains unverified.

The primary demonstration should take approximately **4-6 minutes**:

1. **Weekly Brief:** four commitments.
2. **Incrementality:** strong launch sales versus portfolio movement.
3. **Reorder Quality:** replenishment need versus evidence for expansion.
4. **Production:** fit the candidate into real constraints.
5. **Cash:** distinguish cost exposure from future payments.
6. **Managed Planning:** explain how ScaleSight maintains this work.

The other five pages remain available for deeper questions.

Suggested opening:

> The launch gives us a reason to look closely. The question is what should change in the next production commitment.

Suggested closing:

> With real inputs, we would maintain this comparison alongside your team and bring back the decisions that deserve attention.

03 Public context & evidence boundaries

2.1 The collaboration is a valid current trigger

Alec’s founder describes **Ice Cream for Breakfast**, developed with Purely Elizabeth, as cereal-milk ice cream with a strawberry shell and granola, available nationwide at select Target stores. The current collaboration page identifies Target exclusivity. Source S01

Planning relevance:

• A new flavor can change the portfolio mix.

• Opening distribution and consumer consumption are different signals.

• The collaboration introduces flavor-specific component exposure.

• A Target-exclusive product cannot automatically receive the same allocation treatment as the incumbent range.

Do not infer: actual sales, launch quantities, demand persistence, cannibalization or collaboration end date.

2.2 Culture Cup is an in-house production commitment

Alec’s historical founder post explicitly describes making Culture Cup at its own production facility, including new equipment, packaging and launch work. Source S02

Planning relevance:

A flavor recommendation must account for:

• Available production time.

• Run sizes.

• Changeovers.

• Components.

• Finished-goods release timing.

• Other products competing for capacity.

Unknown: current equipment, actual output rates, yields, shift patterns and production capacity.

2.3 The current business is retail-focused

The current product FAQ says Alec’s has stopped direct website ice-cream delivery and now emphasizes retail availability and grocery delivery partners. Source S03

Demo consequence:

Do not invent an active website subscription or DTC channel.

The core operating channels will be:

• Target.

• Whole Foods.

• Wegmans.

• A separately labeled hypothetical club expansion scenario.

2.4 Ingredient sourcing creates a defensible planning dimension

The current FAQ describes a move away from certified-organic sourcing, partly because limited certified-organic A2 dairy supply previously caused shortages. It says premium A2 dairy remains part of the product. Source S03

Demo consequence:

Use **A2 dairy-based production mix** as a component category.

Do not present:

• A current real dairy shortage.

• The whole current range as certified organic.

• Historical sourcing difficulties as an ongoing verified problem.

2.5 Finance is relevant, but the hiring signal is dated

Alec’s public VP Finance post discusses forecasting, cash, margin, reporting and systems. It supports the relevance of connecting operational decisions to finance; it does not establish that the role remains open. Source S04

The **$11M Series A announcement was published on 9 October 2025**. Its distribution figures are historical, not verified current door counts. Source S05

2.6 Public-data precedence

Use this order:

1. Current rendered product page.

2. Current collection and collaboration pages.

3. Dated founder statements.

4. Dated company announcements.

5. Older commerce metadata.

Some product JSON descriptions contain stale or mismatched copy. Do not automatically use those descriptions as the current ingredient truth.

Unknown internal facts remain unknown: ERP, forecasting software, planning cadence, actual margins, supplier terms, inventory and factory constraints.

Evidence classifications

Every dataset and derived result must carry one of these classifications.

Classification	What belongs here
PUBLIC FACT	Product identity, publicly described ingredients, current collaboration availability
SYNTHETIC INPUT	Sales, inventory, components, costs, production slots, supplier terms
DERIVED METRIC	Cover, shortage, run hours, cash payments, forecast errors
ANALYST INTERPRETATION	Recommendation, confidence assessment, question requiring client judgment

Required metadata:

`evidenceClass`

```
sourceId
asOfDate
snapshotId
assumptionId
note
```

Persistent statement:

> **Illustrative planning concept. Operating data and recommendations are synthetic. No live connection or client results.**

We can show considerable analytical depth while keeping these distinctions explicit.

04.01 Phase 1 / Canonical timeline and units

Field	Canonical setting
Synthetic planning date	12 October 2026
Historical actuals through	11 October 2026
Historical coverage	20 complete weeks
Historical week starts	25 May-5 October 2026
Synthetic launch week	14 September 2026
Primary planning horizon	8 weeks
Extended visibility	13 weeks
W1	12-18 October
W2	19-25 October
W3	26 October-1 November
W4	2-8 November
W5	9-15 November
W6	16-22 November
W7	23-29 November
W8	30 November-6 December
Currency	USD
Base analytical unit	Individual cup equivalent
Retail four-pack	4 cups
Synthetic logistics case	6 four-packs = 24 cups
Review cadence	Weekly, illustrative
Actual client planning cadence	Unknown

The synthetic launch date is not a claim about the real launch date.

Retail POS must be stored as integer four-packs, with cup-equivalent reporting derived by multiplying by four.

Forecasts may contain fractional expected cup equivalents. Booked orders, stock and actual movements must respect their physical pack units.

04.02 Phase 1 / Canonical public product catalog

The current Culture Cup collection supports the following eight products. Source S06

Demo SKU	Public product name	Short planning name
CC01	Alec’s x Purely Elizabeth	Breakfast
CC02	Cookies & Cream	Cookies
CC03	Dark Chocolate Honeycomb Crunch	Honeycomb
CC04	Chocolate Covered Strawberry	Strawberry
CC05	Mint Chocolate Cookie	Mint
CC06	GROUNDWORK Coffee	Coffee
CC07	Madagascar Vanilla Bean	Vanilla
CC08	Peanut Butter Cup	Peanut Butter

These IDs are synthetic, not actual Alec’s SKU codes or GTINs.

Product context for the planning BOM

SKU	Public component context	Synthetic planning categories
CC01	Cereal-milk ice cream, strawberry shell, granola	Base, strawberry shell, granola
CC02	Vanilla ice cream, chocolate cookies, creme shell	Base, creme shell, cookie inclusion
CC03	Chocolate ice cream, honeycomb pieces, chocolatey shell	Base, dark shell, honeycomb inclusion
CC04	Strawberry ice cream, chocolatey shell	Base, dark shell, strawberry preparation
CC05	Mint ice cream, cookies, chocolatey shell	Base, dark shell, cookie/mint ingredients
CC06	Coffee ice cream, chocolatey shell	Base, dark shell, coffee preparation
CC07	Vanilla ice cream, chocolatey shell	Base, dark shell, vanilla preparation
CC08	Peanut-butter ice cream, chocolatey shell	Base, dark shell, peanut-butter preparation

The public pages support these categories; **all quantities and purchasing units below are invented planning assumptions, not recipes**. Source S07

Public product paths:

```
CC01 /products/alecs-x-purely-elizabeth
CC02 /products/culture-cup-cookies-cream-4-pack
CC03 /products/dark-chocolate-honeycomb-crunch
CC04 /products/chocolate-covered-strawberry-4-pack
CC05 /products/mint-chocolate-cookie-4-pack
```

```
CC06 /products/groundwork-coffee-4-pack
CC07 /products/madagascar-vanilla-bean-4-pack
CC08 /products/peanut-butter-cup-4-pack
```

04.03 Phase 1 / Operating scope and retailer eligibility

The modeled business is a **synthetic planning sample**, not Alec's whole company.

Dataset population	Stores	Included in operating demand?
Target launch panel	120	Yes
Whole Foods panel	60	Yes
Wegmans panel	40	Yes
Separate Target comparison panel	60	No - counterfactual research only
Total synthetic stores	**280**	**220 operating + 60 comparison**

IDs:

```
Target operating: T001-T120
Target comparison: TC001-TC060
Whole Foods: WF001-WF060
Wegmans: WG001-WG040
```

No real store addresses or actual store-level results are implied.

SKU eligibility

SKU	Target	Whole Foods	Wegmans	Hypothetical club
CC01	Yes	No	No	No
CC02-CC08	Yes	Yes	Yes	Requires separate pack definition

Only CC01's Target restriction is supported here by public collaboration information. Incumbent retailer × SKU permissions are **synthetic fixture assumptions**.

Do not interpret these as a verified real assortment matrix.

04.04 Phase 1 / SKU × retailer demand dataset

Latest synthetic weekly consumer POS

All figures below are cup equivalents.

SKU	Target panel	Whole Foods	Wegmans	Total operating panel
CC01	2,400	0	0	2,400
CC02	2,400	400	200	3,000
CC03	1,600	280	120	2,000
CC04	1,400	400	200	2,000
CC05	1,000	120	80	1,200
CC06	800	120	80	1,000
CC07	900	200	100	1,200
CC08	600	120	80	800
Total	**11,100**	**1,640**	**860**	**13,600**

Prelaunch comparison

Measure	Prelaunch	Latest	Movement
Target incumbent portfolio	10,000	8,700	-1,300
Target collaboration	0	2,400	+2,400
Total Target panel	10,000	11,100	+1,100 / +11%
Other operating retailers	2,500	2,500	Flat
Total operating panel	12,500	13,600	+1,100 / +8.8%

This distinction matters:

> The new SKU sells 2,400 cups, but the observed operating portfolio grows by 1,100.

It does **not** establish that 1,300 cups were causally cannibalized.

04.05 Phase 1 / Historical SKU dataset

Final eight historical weeks

These are complete canonical Target-panel observations.

Canonical numerical matrix

Week starting	CC01	CC02	CC03	CC04	CC05	CC06	CC07	CC08	Total
17 Aug	0	2,600	1,900	1,600	1,300	1,000	900	700	10,000
24 Aug	0	2,652	1,936	1,632	1,328	1,020	916	716	10,200
31 Aug	0	2,548	1,864	1,568	1,272	980	884	684	9,800
7 Sep	0	2,600	1,900	1,600	1,300	1,000	900	700	10,000
14 Sep	3,200	2,500	1,600	1,400	1,000	800	900	600	12,000
21 Sep	2,800	2,400	1,600	1,400	1,000	800	900	600	11,500
28 Sep	2,500	2,400	1,600	1,400	1,000	800	900	600	11,200
5 Oct	2,400	2,400	1,600	1,400	1,000	800	900	600	11,100

The four prelaunch weeks average exactly **10,000 cups**.

The four postlaunch weeks total **45,800 cups**, including **10,900 collaboration cups**.

Earlier twelve weeks

Use the following total-volume multipliers against the 10,000-cup prelaunch SKU mix:

Week starting	Multiplier	Target total
25 May	0.94	9,400
1 Jun	0.96	9,600
8 Jun	0.98	9,800
15 Jun	1.00	10,000
22 Jun	1.02	10,200
29 Jun	1.04	10,400
6 Jul	1.00	10,000
13 Jul	0.98	9,800
20 Jul	1.02	10,200
27 Jul	1.00	10,000
3 Aug	1.04	10,400
10 Aug	1.02	10,200

Deterministic allocation rule:

1. Convert the weekly total into four-packs.
2. Allocate packs according to the prelaunch SKU shares.
3. Use largest-remainder rounding.
4. Multiply back by four.
5. Preserve the weekly total exactly.

Whole Foods and Wegmans use their retailer × SKU quantities from Section 7 throughout this deliberately controlled historical fixture.

That holds non-Target demand constant while investigating the launch panel.

04.06 Phase 1 / Store-cohort dataset

Four cohorts of 30 Target operating stores.

Cohort	Stores	Synthetic interpretation to investigate
A	T001-T030	Strong launch and stronger portfolio movement
B	T031-T060	Strong launch but flat portfolio movement
C	T061-T090	Weaker launch; incumbents remain more important
D	T091-T120	Launch demand persists through later order cycles

These are **defined synthetic cohorts**, not proof of trained ML clustering.

Prelaunch SKU mix by cohort

SKU	A	B	C	D	Total
CC02	648	652	648	652	2,600
CC03	476	476	476	472	1,900
CC04	400	400	400	400	1,600
CC05	324	324	324	328	1,300
CC06	252	248	252	248	1,000
CC07	224	224	224	228	900
CC08	176	176	176	172	700
Total	**2,500**	**2,500**	**2,500**	**2,500**	**10,000**

Latest SKU mix by cohort

SKU	A	B	C	D	Total
CC01	900	800	200	500	2,400
CC02	700	400	748	552	2,400
CC03	452	300	400	448	1,600
CC04	352	248	352	448	1,400
CC05	248	200	252	300	1,000
CC06	200	152	248	200	800
CC07	248	248	200	204	900
CC08	200	152	100	148	600
Total	**3,300**	**2,500**	**2,500**	**2,800**	**11,100**

This is the core contrast:

• A: 900 launch cups accompany 800 cups of observed portfolio growth.

• B: 800 launch cups accompany no observed portfolio growth.

• C: 200 launch cups accompany no observed portfolio growth.

• D: 500 launch cups accompany 300 cups of observed portfolio growth.

None of those observations alone proves causation.

Availability data

For the latest week:

Cohort	Possible store-days	Available store-days	Availability	CC01 cups per available store-week
A	210	200	95.2%	31.50
B	210	196	93.3%	28.57
C	210	168	80.0%	8.33
D	210	204	97.1%	17.16

Formula:

```
Equivalent available store-weeks = available store-days / 7

Availability-adjusted observed velocity =
POS cups / equivalent available store-weeks
```

This normalizes observed selling exposure. It does not automatically recover all lost demand.

Individual-store generation

For each cohort:

- Allocate integer POS four-packs across its 30 stores.
- Use fixed store weights derived from seed 241.
- Ensure allocations sum exactly to the cohort totals.
- Assign available days consistent with the aggregate table.
- Keep store weights fixed through time so store comparisons are meaningful.
- Store `listed`, `availableDays`, `promoFlag`, `priceIndex` and `retailerReceipts` separately.

The dense historical table contains:

```
280 stores × 8 SKUs × 20 weeks = 44,800 rows
```

Nonlisted combinations receive zero POS and `listed=false`.

Comparison-panel rows are excluded from operating totals.

04.07 Phase 1 / Counterfactual dataset

The comparison panel consists of four groups of 15 nonlaunch Target stores.

Their prelaunch portfolio baseline is 1,000 cups per group.

Matched group	Prelaunch control cups	Latest control cups	Control movement	Launch-panel counterfactual
A	1,000	1,040	+4%	2,600
B	1,000	1,020	+2%	2,550
C	1,000	980	-2%	2,450
D	1,000	1,080	+8%	2,700
Total	**4,000**	**4,120**	**+3%**	**10,300**

For this controlled fixture, comparison-group selling exposure is proportionately matched to the corresponding launch cohort.

Calculation:

```
Counterfactual cohort demand =
prelaunch launch-cohort demand
× latest control demand / prelaunch control demand
```

Estimated portfolio movement against that counterfactual

Cohort	Observed portfolio	Counterfactual	Estimated difference
A	3,300	2,600	+700
B	2,500	2,550	-50
C	2,500	2,450	+50
D	2,800	2,700	+100
Total	**11,100**	**10,300**	**+800**

Interpretation:

> Under this illustrative comparison method, portfolio demand is 800 cups above the modeled no-launch case.

Use a sensitivity range of ±500 cups around the counterfactual:

```
Counterfactual range: 9,800-10,800
Associated portfolio difference: 300-1,300 cups
```

This is an **assumption sensitivity range**, not a statistical confidence interval.

Two ratios can be calculated, but must remain distinct:

Metric	Calculation	Meaning
Observed net-growth ratio	1,100 / 2,400 = 45.8%	Descriptive comparison
Counterfactual-associated lift ratio	800 / 2,400 = 33.3%	Estimate under stated assumptions

Do not label either as proven customer-level incrementality.

What a real investigation would additionally require: comparable distribution, availability, pricing, promotion, seasonality and - where accessible - customer purchasing evidence.

04.08 Phase 1 / Launch fill, reorders and retailer inventory

Collaboration POS by cohort over the launch period

Launch week	A	B	C	D	Total
14 Sep	1,200	1,100	300	600	3,200
21 Sep	1,048	952	248	552	2,800
28 Sep	948	852	220	480	2,500
5 Oct	900	800	200	500	2,400
Cumulative	**4,096**	**3,704**	**968**	**2,132**	**10,900**

Retail receipt records

These quantities are case-compatible: multiples of 24 cups.

Receipt ID	Cohort	Type	Received week	Cups
LF-A	A	Initial fill	14 Sep	2,880
LF-B	B	Initial fill	14 Sep	2,880
LF-C	C	Initial fill	14 Sep	1,200
LF-D	D	Initial fill	14 Sep	1,920
R-A1	A	Reorder 1	21 Sep	1,440
R-A2	A	Reorder 2	28 Sep	1,200
R-A3	A	Reorder 3	5 Oct	960
R-B1	B	Reorder 1	21 Sep	960
R-D1	D	Reorder 1	28 Sep	720
R-D2	D	Reorder 2	5 Oct	720

Assume zero collaboration retailer inventory before initial fill.

Reconciled collaboration retailer inventory

Cohort	Cumulative receipts	Cumulative POS	Closing retailer inventory
A	6,480	4,096	2,384
B	3,840	3,704	136
C	1,200	968	232
D	3,360	2,132	1,228
Total	**14,880**	**10,900**	**3,980**

Formula:

```
Retail closing inventory =
retail opening inventory + receipts - consumer POS - write-offs
```

Important conclusions:

• A has stronger portfolio movement and several receipt cycles.

• B has strong launch sales but little remaining retailer inventory and flat portfolio movement.

• C has weaker launch sales and lower availability.

• D has persistent launch consumption and later receipt cycles.

B may still need replenishment even if its launch demand is mostly a mix change. Service needs and incrementality are different questions.

These 3,980 retailer-owned cups are **not included** in the brand-owned finished-goods stock below.

04.09 Phase 1 / Forward demand and dispatch assumptions

The inventory engine needs a forecast of **brand dispatch requirements**, not a direct subtraction of consumer POS from factory inventory.

Use:

```
Required retailer receipts =
forecast consumer demand
+ desired ending retailer inventory
- opening retailer inventory
```

In the base fixture:

• Retailer inventory targets remain unchanged.

• Aggregate dispatch requirements therefore match the selected consumer-demand forecast.

• Order-timing scenarios can change dispatch timing without changing consumer demand.

• Firm orders consume the dispatch forecast; they are not added a second time.

Central eight-week dispatch forecast

SKU	W1	W2	W3	W4	W5	W6	W7	W8	Eight-week total
CC01	2,400	2,300	2,200	2,000	1,800	1,700	1,600	1,600	15,600
CC02	3,000	3,000	3,000	3,000	3,000	3,000	3,000	3,000	24,000
CC03	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	16,000
CC04	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	16,000
CC05	1,200	1,200	1,200	1,200	1,200	1,200	1,200	1,200	9,600
CC06	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	8,000
CC07	1,200	1,200	1,200	1,200	1,200	1,200	1,200	1,200	9,600
CC08	800	800	800	800	800	800	800	800	6,400
Total	**13,600**	**13,500**	**13,400**	**13,200**	**13,000**	**12,900**	**12,800**	**12,800**	**105,200**

The collaboration’s declining central forecast is a **synthetic planning assumption**, not a claim that the real launch is fading.

Extended horizon

For W9-W13:

• CC01 remains at 1,600 cups/week.

• Other SKUs retain their central weekly quantities.

• No additional production receipts are silently invented.

• Resulting shortages identify the next commitment that needs review.

Sensitivity assumptions

Product group	Lower case	Central	Higher case
Collaboration	75% of central	100%	125%
Incumbents	90% of central	100%	115%

These are scenario bands, not calibrated probabilistic intervals.

04.10 Phase 1 / Canonical finished-goods position

All stock is synthetic, saleable and brand-owned.

SKU	Opening stock	W1 dispatch	Simple opening cover	Buffer policy
CC01	4,800	2,400	2.0 weeks	1.5 forward weeks
CC02	6,000	3,000	2.0	2.0
CC03	4,800	2,000	2.4	2.0
CC04	6,000	2,000	3.0	2.0
CC05	7,200	1,200	6.0	2.0
CC06	4,800	1,000	4.8	2.0
CC07	3,600	1,200	3.0	2.0
CC08	4,800	800	6.0	2.0
Total	**42,000**	**13,600**	-	-

Buffer policies are illustrative client-policy assumptions.

Use forward demand to assess future buffer breaches. Do not apply today’s run rate indefinitely when the forecast changes.

Inventory lot records

Use two lot records per SKU:

• Older lot: 40% of opening stock.

• Newer lot: 60%.

• Both remain saleable throughout the default horizon.

• Synthetic expiry dates: 28 February and 31 March 2027.

• Allocate oldest eligible stock first.

• Store quantity, owner, location, pack format, expiry and QA status.

The quantities reconcile exactly because all opening balances are divisible by five.

No actual Alec’s shelf life or expiry date is implied.

04.11 Phase 1 / Production commitments

Next production-window alternatives

SKU	Requested cups	Reviewed cups	Change
CC01	12,000	8,000	-4,000
CC02	10,000	12,000	+2,000
CC03	6,000	4,000	-2,000
CC04	6,000	6,000	0
CC05	4,000	2,000	-2,000
CC06	4,000	2,000	-2,000
CC07	6,000	4,000	-2,000
CC08	4,000	2,000	-2,000
Total	**52,000**	**40,000**	**-12,000**

Production-window assumptions

Parameter	Synthetic value
Available line hours	40
Output rate	1,200 good cups/hour
Minimum nonzero run	2,000 cups
Quantity increment	2,000 cups
Changeover between flavors	0.5 hour
Reviewed active flavors	8
Reviewed changeovers	7
Production period	19-23 October
Finished-goods availability	Start of W3, 26 October
Generic packaging available	40,000 cup/lid sets
Base mix available	3,200 kg

“Good cups/hour” already incorporates the illustrative yield allowance. Do not apply yield twice.

The 40 hours are the time allocated to this modeled window after other business requirements - not a claim about the whole factory’s weekly capacity.

Capacity calculation

```
Run hours = total good cups / 1,200

Changeover hours = 0.5 × (active flavor runs - 1)

Total required hours = run hours + changeover hours
```

Option	Run hours	Changeovers	Total hours	Feasible?
Requested 52,000	43.33	3.50	46.83	No
Reviewed 40,000	33.33	3.50	36.83	Yes

The reviewed option leaves **3.17 line hours**, but consumes the available generic packaging and base mix.

Therefore:

> Spare line time does not automatically mean spare production capability.

04.12 Phase 1 / Later production receipts

The projection includes later supply commitments as well as the first run.

Synthetic already-planned bridge commitment

SKU	Bridge quantity	Availability
CC01	4,000	W6
CC02	10,000	W6
CC03	2,000	W5
CC03	6,000	W6
CC04	4,000	W6
CC05	4,000	W6
CC06	2,000	W6
CC07	6,000	W6
CC08	2,000	W6
Total	**40,000**	-

The early CC03 tranche matters:

> Without the 2,000 cups in W5, Honeycomb has a modeled 1,200-cup shortage before the later receipt.

This is a secondary resequencing dependency behind the four main decisions.

Bridge production uses:

```
40,000 / 1,200 + 8 × 0.5 = 37.33 hours
```

Nine runs, because CC03 is produced in two tranches.

Assume 40 reserved line hours across the bridge production period. This is a separate future capacity reservation.

Commitment statuses

Commitment	Status in fixture
Requested 52,000-cup alternative	Uncommitted and infeasible
Reviewed 40,000-cup alternative	Proposed for client decision
Bridge 40,000 cups	Synthetic confirmed planning commitment
Early Honeycomb tranche	Synthetic confirmed dependency

None represents a real approval by Alec’s.

04.13 Phase 1 / Reconciled inventory trajectory

This projection assumes:

• Reviewed 40,000-cup run.

• Bridge receipts above.

• Central dispatch forecast.

• Feasible node allocation.

• No write-offs.

• Receipts available at the start of the specified week.

End-of-week finished-goods stock

SKU	W1	W2	W3	W4	W5	W6	W7	W8
CC01	2,400	100	5,900	3,900	2,100	4,400	2,800	1,200
CC02	3,000	0	9,000	6,000	3,000	10,000	7,000	4,000
CC03	2,800	800	2,800	800	800	4,800	2,800	800
CC04	4,000	2,000	6,000	4,000	2,000	4,000	2,000	0
CC05	6,000	4,800	5,600	4,400	3,200	6,000	4,800	3,600
CC06	3,800	2,800	3,800	2,800	1,800	2,800	1,800	800
CC07	2,400	1,200	4,000	2,800	1,600	6,400	5,200	4,000
CC08	4,000	3,200	4,400	3,600	2,800	4,000	3,200	2,400

Reconciliation:

```
Opening stock                  42,000
Reviewed production            40,000
Bridge receipts                40,000
Total available               122,000
Eight-week served demand      105,200
Closing stock                  16,800
```

Zero aggregate unmet demand does not mean zero risk.

The projection includes:

• Very little collaboration stock before W3.

• Cookies reaching zero at the end of W2.

• Several buffer breaches.

• Strawberry reaching zero at W8.

• Dependence on the early Honeycomb bridge tranche.

• Regional allocation risk.

If the reviewed run is not committed

Retain only opening stock and bridge receipts.

SKU	Eight-week modeled unmet dispatch
CC01	6,800
CC02	9,000
CC03	3,200
CC04	6,000
CC05	0
CC06	1,200
CC07	2,400
CC08	0
Total	**28,600**

Some stock remains in the wrong products or arrives after a shortage.

That is why a portfolio-level inventory total is not enough to make the decision.

04.14 Phase 1 / Planning BOM

Common components per good cup

Component	Synthetic coefficient	Cost assumption
Dairy-based production mix	0.080 kg	$3.50/kg
Cup/lid set	1 set	$0.10/set
Printed four-pack carton	0.25 carton	$0.08/carton
Culture preparation	0.00005 kg	$500/kg
Fiber preparation	0.002 kg	$5/kg

These common components contribute **$0.435 per cup** in modeled material cost.

The base mix is a planning component that already includes common base ingredients. Do not additionally add its internal milk, cream, sugar and egg constituents and double-count them.

Flavor-specific components

SKU	Shell coefficient	Inclusion coefficient	Other ingredient cost per cup
CC01	Strawberry shell 0.015 kg	Granola 0.012 kg	$0.085
CC02	Creme shell 0.015 kg	Cookie 0.010 kg	$0.005
CC03	Dark shell 0.015 kg	Honeycomb 0.010 kg	$0.065
CC04	Dark shell 0.015 kg	Strawberry preparation 0.010 kg	$0.085
CC05	Dark shell 0.015 kg	Cookie 0.010 kg	$0.035
CC06	Dark shell 0.015 kg	Coffee preparation 0.003 kg	$0.065
CC07	Dark shell 0.015 kg	Vanilla preparation 0.001 kg	$0.025
CC08	Dark shell 0.015 kg	Peanut preparation 0.012 kg	$0.065

“Other ingredients” are separate SKU-specific synthetic kits, with 20,000 cup-equivalents available per SKU.

They are not interchangeable between flavors.

Flavor-component prices

Component	Synthetic price
Strawberry shell	$8/kg
Creme shell	$8/kg
Dark shell	$6/kg
Granola	$5/kg
Cookie inclusion	$4/kg
Honeycomb inclusion	$6/kg
Strawberry preparation	$4/kg
Coffee preparation	$20/kg
Vanilla preparation	$50/kg
Peanut preparation	$5/kg

These coefficients support planning calculations only. They are not food formulation, quality or nutrition specifications.

04.15 Phase 1 / Component availability dataset

Available before the next production window

Component	Available quantity	Reviewed requirement	Remaining after reviewed run
Base mix	3,200 kg	3,200 kg	0
Cup/lid sets	40,000	40,000	0
Culture preparation	2 kg	2 kg	0
Fiber preparation	80 kg	80 kg	0
Strawberry shell	150 kg	120 kg	30 kg
Creme shell	220 kg	180 kg	40 kg
Dark shell	420 kg	300 kg	120 kg
Granola	120 kg	96 kg	24 kg
Cookie inclusion	180 kg	140 kg	40 kg
Honeycomb inclusion	60 kg	40 kg	20 kg
Strawberry preparation	90 kg	60 kg	30 kg
Coffee preparation	30 kg	6 kg	24 kg
Vanilla preparation	12 kg	4 kg	8 kg
Peanut preparation	60 kg	24 kg	36 kg

Printed carton stock

SKU	Available cartons	Reviewed requirement
CC01	3,000	2,000
CC02	3,500	3,000
CC03	1,500	1,000
CC04	1,800	1,500
CC05	1,500	500
CC06	1,500	500
CC07	1,500	1,000
CC08	1,200	500

Collaboration depth comparison

CC01 output	Granola requirement	Strawberry shell requirement
8,000 cups	96 kg	120 kg
10,000 cups	120 kg	150 kg
12,000 cups	144 kg	180 kg

At 12,000 cups:

• Granola is short by 24 kg.

• Strawberry shell is short by 30 kg.

• Portfolio-level base, packaging and line constraints must still be resolved.

This makes the recommendation more useful than “the forecast increased.”

04.16 Phase 1 / Component purchase orders

These are synthetic supplier commitments supporting the bridge production.

Purchase-order headers

PO	Contents	Receipt date	Order value	Previously paid	Future balance
AP-101	Base mix	2 Nov	$11,200	$3,360	$7,840
AP-102	Cup/lid sets	2 Nov	$4,000	$1,200	$2,800
AP-103	Printed cartons	2 Nov	$800	$240	$560
AP-104	Culture and fiber	2 Nov	$1,400	$420	$980
AP-105	Flavor components	2 Nov	$5,550	$1,665	$3,885
Total	-	-	**$22,950**	**$6,885**	**$16,065**

Terms:

- 30% paid before the snapshot.
- 70% due on receipt.
- Supplier identities are invented.
- Standard lead-time assumption: three weeks.
- Do not claim actual supplier performance.

PO line items

```
AP-101
BASE: 3,200 kg × $3.50

AP-102
CUP-LID: 40,000 sets × $0.10

AP-103
CC01 cartons: 1,000
CC02 cartons: 2,500
CC03 cartons: 2,000
CC04 cartons: 1,000
CC05 cartons: 1,000
CC06 cartons: 500
CC07 cartons: 1,500
CC08 cartons: 500
Total: 10,000 cartons × $0.08

AP-104
Culture preparation: 2 kg × $500
Fiber preparation: 80 kg × $5

AP-105
Granola: 60 kg × $5
Dark shell: 300 kg × $6
Creme shell: 150 kg × $8
Strawberry shell: 60 kg × $8
Cookie inclusion: 100 kg × $4
Honeycomb inclusion: 80 kg × $6
Strawberry preparation: 60 kg × $4
Coffee preparation: 10 kg × $20
Vanilla preparation: 6 kg × $50
Peanut preparation: 30 kg × $5
```

All receipts arrive before the early Honeycomb bridge production.

Optional extra component reservation

Separate from AP-105:

Field	Value
PO	AP-106
Component	Granola
Quantity	60 kg
Value	$300
Deposit	$90
Future balance	$210
Earliest standard receipt	26 October
Cancellation/refund terms	Unknown
Status	Uncommitted option

Reserving this component:

- Creates a financial commitment.
- Does not automatically reserve factory capacity.
- Does not authorize more finished-goods output.
- Must not be described as cost-free or fully reversible.

04.17 Phase 1 / Retailer × SKU allocation dataset

The main allocation story uses **CC04, Chocolate Covered Strawberry**.

Brand-owned stock locations

Location	CC04 stock	W1-W2 demand	Additional protected buffer	Transferable surplus
Target West pool	600	1,800	0	0
Target East pool	3,600	1,000	680	1,920
Central pool for other retailers	1,800	1,200	0	600
Total	**6,000**	**4,000**	**680**	-

Target West demand is 900 cups/week.

Target East demand is 500 cups/week.

Together they reconcile to the Target CC04 forecast of 1,400 cups/week.

Whole Foods and Wegmans require the remaining 600 cups/week.

Transfer option

```
Transfer 1,200 CC04 cups
Target East pool → Target West pool
```

Measure	Before	After
West stock available for two-week demand	600	1,800
West modeled shortage	1,200	0
East stock	3,600	2,400
East transferable surplus after protection	1,920	720
Total network stock	6,000	6,000

Synthetic execution assumptions:

- Same finished SKU and four-pack format.
- 50 logistics cases.
- Two-day frozen transit.
- One day for receiving/release.
- Freight $320.
- Handling/QA $80.
- Total transfer expense $400.

• Stock available before West’s projected depletion.

These are **brand-owned pools**, not retailer-owned stock sitting on store shelves.

Eligibility rules

A transfer is feasible only if:

```
same finished SKU
AND compatible pack
AND destination permitted
AND donor protection remains satisfied
AND arrival precedes need
AND lot remains saleable
AND cold-chain conditions can be maintained
```

Do not convert one flavor into another.

Do not transfer CC01 into non-Target channels.

Do not treat a four-pack as a club pack.

04.18 Phase 1 / SKU economics and cash

This section deliberately separates **production cost** from **cash still to be paid**.

Unit-cost assumptions

SKU	Material cost/cup	Conversion cost/cup	Total production cost/cup
CC01	$0.70	$0.50	$1.20
CC02	$0.60	$0.45	$1.05
CC03	$0.65	$0.45	$1.10
CC04	$0.65	$0.50	$1.15
CC05	$0.60	$0.40	$1.00
CC06	$0.65	$0.45	$1.10
CC07	$0.60	$0.35	$0.95
CC08	$0.65	$0.45	$1.10

Conversion represents synthetic incremental production labor, utilities and other conversion expenditure.

It is not a supplier’s finished-goods deposit: Alec’s makes Culture Cup in-house.

Production-cost comparison

Measure	Requested 52,000	Reviewed 40,000
Materials consumed at modeled cost	$33,400	$25,500
Conversion cost	$23,500	$18,200
Total production cost	**$56,900**	**$43,700**

Difference:

```
$56,900 - $43,700 = $13,200
```

Label:

$13,200 lower modeled production-cost exposure.

Not:

“ScaleSight saved Alec’s $13,200.”

Future cash for the reviewed run

For this fixture:

- Materials already on hand were paid for before the snapshot.
- No additional material purchase is needed for the reviewed first window.
- Future conversion cash is $18,200.

• The regional transfer adds $400.

Therefore:

```
Reviewed first-window future cash = $18,600
```

Synthetic first-window cash limit: **$25,000**.

Remaining room:

```
$25,000 - $18,600 = $6,400
```

Requested-plan material top-ups

The larger request requires additional materials.

Purchase	Synthetic order quantity	Cash value
Base mix	1,000 kg	$3,500
Cup/lid sets	12,000	$1,200
Culture preparation	1 kg	$500
Fiber preparation	25 kg	$125
Granola	60 kg	$300
Dark shell	60 kg	$360
Strawberry shell	60 kg	$480
Total	-	**$6,465**

The order quantities reflect synthetic purchasing multiples, not just exact shortages.

The requested alternative would therefore require at least:

```
Conversion cash      $23,500
Material top-ups       $6,465
Total                 $29,965
```

This excludes any expedited delivery, overtime or extra capacity cost.

It exceeds the first-window cash limit and remains infeasible on timing and capacity.

Eight-week cash schedule

Date	Payment	Amount
23 Oct	60% of reviewed conversion expenditure	$10,920
26 Oct	CC04 transfer expense	$400
30 Oct	Remaining reviewed conversion expenditure	$7,280
2 Nov	Existing component-PO balances	$16,065
13 Nov	60% of bridge conversion expenditure	$10,560
20 Nov	Remaining bridge conversion expenditure	$7,040
Total future cash	-	**$52,265**

Bridge conversion cost is **$17,600**.

Previously paid PO deposits of $6,885 are excluded from future cash.

Never add material consumption cost and the corresponding PO payments as two separate cash outflows.

Revenue-exposure assumptions

Use synthetic brand net revenue per delivered cup:

SKU	Net revenue/cup
CC01	$1.90
CC02	$1.80
CC03	$1.85
CC04	$1.80
CC05	$1.75
CC06	$1.85
CC07	$1.70
CC08	$1.85

This is not shelf price.

```
Modeled unserved-dispatch revenue exposure =
unserved cups × synthetic net revenue/cup
```

It is an exposure estimate - not guaranteed lost revenue, profit or sales recovered.

04.19 Phase 1 / Forecast-learning dataset

The demo should make analytical capability visible without pretending we have trained an Alec’s model on private data.

Use two layers.

Layer A - reproducible analytical methods

Implement:

• Availability-normalized observed velocity.

• Retailer/SKU pooling for sparse observations.

• Separate launch-age treatment.

• Forecast reconciliation across SKU, retailer and portfolio.

• Counterfactual comparison under explicit assumptions.

• Stock and component simulation.

• Baseline-versus-reviewed forecast evaluation.

Layer B - illustrative historical forecast ledger

The following forecasts are synthetic archived forecasts, treated as having been issued before the corresponding outcome.

They are **not evidence of a trained real-world ML model**.

SKU	Actuals over four launch weeks	Naive forecast path	Reviewed illustrative path
CC01	3200, 2800, 2500, 2400	2800, 3200, 2800, 2500	2800, 2900, 2600, 2450
CC02	2500, 2400, 2400, 2400	2600, 2500, 2400, 2400	2480, 2440, 2400, 2400
CC03	1600, 1600, 1600, 1600	1900, 1600, 1600, 1600	1760, 1640, 1600, 1600
CC04	1400, 1400, 1400, 1400	1600, 1400, 1400, 1400	1520, 1440, 1400, 1400
CC05	1000, 1000, 1000, 1000	1300, 1000, 1000, 1000	1120, 1040, 1000, 1000
CC06	800, 800, 800, 800	1000, 800, 800, 800	920, 840, 800, 800
CC07	900, 900, 900, 900	900, 900, 900, 900	900, 900, 900, 900
CC08	600, 600, 600, 600	700, 600, 600, 600	640, 620, 600, 600

Calculated benchmark results

Metric	Naive path	Reviewed illustrative path
Total actual cups	45,800	45,800
Total absolute error	2,500	1,450
WAPE	5.46%	3.17%
Signed forecast error	+1,700	+610
Bias percentage	+3.71%	+1.33%

These percentages are derived from the supplied synthetic ledger.

They must not become “our forecasting accuracy” claims.

Methods to specify for implementation or a real pilot

Method	Purpose	Required treatment
Hierarchical forecasting	Stabilize sparse SKU/store forecasts	Reconcile totals; compare against a simple baseline
Launch-decay estimation	Separate opening excitement from persistence	Require analogous launches or sufficient history
Counterfactual modeling	Estimate portfolio outcome without launch	State assumptions and confounders
Store clustering	Find similar commercial patterns	Label defined cohorts versus fitted clusters
Anomaly detection	Identify signals needing investigation	Explain the trigger and data quality
Scenario simulation	Trace demand changes into commitments	Apply supply and cash constraints

Do not invent:

- A repeat-purchase probability.
- Feature-importance percentages.
- A live training status.
- A confidence score such as “94%.”
- A guaranteed uplift from ML.

04.20 Phase 1 / Scenario dataset

Each scenario changes specific inputs, then reruns the same inventory and constraint engine.

Scenario register

ID	Change	What remains unchanged
S0	Reviewed baseline	Canonical data
S1	CC01 demand -20%	Production receipts
S2	CC01 demand +25%	Production receipts
S3	CC02 demand +20%	Production receipts
S4	Move CC01 W2 dispatch requirement into W1	Total horizon demand
S5	Base-mix availability -20%	Packaging and demand
S6	Hypothetical 4,000-cup club launch	Existing pack eligibility
S7	Reviewed run released one week late	Quantity and demand

Calculated eight-week outcomes before corrective action

Scenario	Modeled unmet dispatch	Closing finished stock	Important interpretation
S0 Baseline	0	16,800	Thin timing/buffer positions remain
S1 Collaboration -20%	0	19,920	CC01 closes at 4,320
S2 Collaboration +25%	2,700	15,600	Shortage occurs before W3 and again at W8
S3 Cookies +20%	1,200	13,200	Shortage before W3
S4 Earlier CC01 dispatch	0	16,800	W1 stock falls to 100; timing changes, total demand does not
S7 Run one week late	6,300	23,100	Later stock does not undo earlier unserved demand

The last scenario demonstrates an important point:

> Ending inventory can rise while service gets worse, because supply arrived after the need.

S2 detail - stronger collaboration demand

CC01 unmet dispatch:

- W2: 1,075 cups.
- W8: 1,625 cups.
- Total: 2,700.

Possible responses to investigate:

1. Bring a small tranche forward.
2. Change the first-window flavor mix.
3. Change the bridge mix.
4. Obtain additional components.
5. Allocate existing supply to confirmed requirements.
6. Wait on discretionary depth while preserving essential supply.

Producing more in W3 alone does not resolve the W2 shortage.

S3 detail - stronger Cookies demand

At 3,600 cups/week:

```
Demand before W3 receipt = 7,200
Opening stock = 6,000
Timing shortfall = 1,200
```

A larger quantity released in W3 still arrives too late.

The decision is initially about **timing or compatible supply**, not only total output.

S5 detail - base mix tightens

Available base:

```
3,200 kg × 80% = 2,560 kg
2,560 / 0.080 = 32,000 cups
```

A candidate constrained mix:

SKU	Candidate cups
CC01	6,000
CC02	12,000
CC03	4,000
CC04	4,000
CC05	0
CC06	0
CC07	4,000
CC08	2,000
Total	**32,000**

This uses six active flavor runs:

```
32,000 / 1,200 + 5 × 0.5 = 29.17 hours
```

It is a feasible manufacturing candidate, **not a commercially optimal recommendation**.

Calculated unmet dispatch over eight weeks:

• CC01: 800.

• CC04: 2,000.

• CC06: 1,200.

• Total: 4,000.

Compare this with obtaining earlier base supply or changing timing. Do not automatically choose the cheapest feasible plan.

S6 detail - hypothetical club launch

Assume:

• 4,000 cups.

• Eight cups per club pack.

• 500 club packs.

• Distinct club carton.

• Separate pack eligibility and packing requirement.

• No approved conversion of existing retail four-packs.

Default outcome:

Blocked pending club-pack supply and production definition.

Do not show a successful allocation from retail four-pack inventory.

The scenario can quantify the additional requirements, but it must not fabricate a feasible execution path.

04.21 Phase 1 / Calculation engine

24.1 Inventory and unmet demand

For each SKU × brand-owned node × period:

```
Available supply =
opening inventory
+ released production
+ incoming finished goods
+ inbound transfers
- outbound transfers
- write-offs
```

Then:

```
Served dispatch = min(required dispatch, available supply)

Unmet dispatch = required dispatch - served dispatch

Closing inventory = available supply - served dispatch
```

Negative stock must never be stored as physical inventory.

Keep:

• Unmet demand.

• Backlogged orders.

• Lost orders.

as separate states. The base scenario treats unmet dispatch as unserved within that period, not automatically backlogged.

24.2 Cover

Simple descriptive cover:

```
Opening cover = opening stock / next-week demand
```

Forward cover:

> Number of future periods that available stock can satisfy using the actual forecast sequence.

Use forward cover for changing launch demand.

24.3 Component demand

```
Component requirement =
sum(production quantity × SKU coefficient)
```

Include only production being evaluated.

Do not charge the reviewed run and the requested alternative simultaneously.

24.4 Manufacturing feasibility

Check:

• Minimum run.

• Quantity increments.

• Active-run changeovers.

- Line hours.
- Component quantities.
- Component receipt dates.
- Printed carton compatibility.
- Release timing.

A quantity-only capacity limit is insufficient.

24.5 Allocation

```
Transferable stock =
eligible stock
- protected dispatch before replenishment
- required donor buffer
```

Transfers conserve network stock.

24.6 Cost and cash

```
Production cost =
material consumption value + conversion cost
```

```
Future cash =
unpaid procurement payments
+ future conversion expenditure
+ incremental logistics/other payments
```

Previously paid materials are not paid again.

24.7 Forecast evaluation

```
WAPE = sum(abs(forecast - actual)) / sum(actual)
```

```
Bias = sum(forecast - actual) / sum(actual)
```

If actual total is zero, return null and explain why.

24.8 Counterfactual movement

```
Estimated associated lift =
observed portfolio - modeled no-launch portfolio
```

The method and uncertainty must accompany the number.

04.22 Phase 1 / Recommendation logic and confidence

Do not generate recommendations from a single threshold such as “sales rose 20%.”

Action rules

Action	Required reasoning
PRODUCE	Projected requirement, evidence and feasible supply support more output
PROTECT CAPACITY	A supported requirement risks displacement by a less-established commitment
HOLD ADDITIONAL DEPTH	Existing stock/commitments cover the central requirement; stronger evidence is needed
REALLOCATE	Eligible surplus can arrive where needed without breaking donor protection
RESEQUENCE	Timing changes solve a gap better than adding total output
RESERVE COMPONENT	Component optionality has value under explicit cost and supplier terms
INVESTIGATE	Conflicting evidence or missing information could materially change the commitment
WATCH	Signal merits monitoring but not an immediate change

Two confidence assessments

Keep them separate:

1. **Demand-evidence confidence.**

2. **Execution-feasibility confidence.**

Examples:

• CC01: moderate demand evidence because it is a recent launch.

• CC02: established demand history, but low timing resilience before W3.

• CC04: aggregate supply is adequate; allocation confidence depends on the transfer assumptions.

• Component reservation: terms remain incomplete even if quantity arithmetic is clear.

Use **Low** / **Moderate** / **High**, with an explanation.

No artificial percentage score.

The following specifies what each of the 11 pages needs from Phase 1. It does not prescribe layout or UI.

05 Data dictionary, relationships & engine contracts

The essential fields below are supplied in Phase 1. Keys, grains, type/null rules and relationship notes are added technical design choices. They do not describe actual Alec's systems. Every record also carries evidenceClass, sourceId, asOfDate, snapshotId, assumptionId and note.

Product

Key: `productId` **Grain:** One public product **Units / classification:** Public identity; technical ID synthetic

Essential fields: `productId, publicName, publicSource, format`

PlanningSKU

Key: `skuId` **Grain:** One planning SKU / pack **Units / classification:** Synthetic; cup and pack units

Essential fields: `skuId, productId, cupsPerPack, casePack, active`

Retailer

Key: `retailerId` **Grain:** One modeled retailer panel **Units / classification:** Public name; panel synthetic

Essential fields: `retailerId, publicName, syntheticPanelScope`

Store

Key: `storeId` **Grain:** One synthetic store **Units / classification:** Synthetic; retailer/cohort FK

Essential fields: `storeId, retailerId, cohortId, operatingOrControl`

StoreWeek

Key: `storeId + skuId + week` **Grain:** Store-SKU-week **Units / classification:** Synthetic; integer four-packs, days

Essential fields: `storeId, skuId, week, packsSold, availableDays, listed, promoFlag, priceIndex`

RetailReceipt

Key: `receiptId` **Grain:** One retail receipt event **Units / classification:** Synthetic; cups and received date

Essential fields: `receiptId, skuId, cohortOrStoreId, type, receivedDate, cups`

RetailInventory

Key: `scopeId + skuId + week` **Grain:** Retail stock balance **Units / classification:** Derived; retailer-owned cups

Essential fields: `storeOrCohortId, skuId, opening, receipts, POS, writeOffs, closing`

BrandInventoryLot

Key: `lotId` **Grain:** One brand-owned lot/node **Units / classification:** Synthetic; cup quantity and expiry

Essential fields: `lotId, skuId, nodeId, quantity, packFormat, expiry, QAStatus, owner`

DispatchForecast

Key: `skuId + retailerId + week + version` **Grain:** Dispatch forecast bucket **Units / classification:** Synthetic input; expected cups

Essential fields: `skuId, retailerId, week, central, low, high, method, version`

FirmOrder

Key: `orderId` **Grain:** One firm order line **Units / classification:** Synthetic if supplied; cups

Essential fields: `orderId, skuId, retailerId, requiredDate, cups, forecastConsumption`

ProductionRun

Key: `runId` **Grain:** One production run **Units / classification:** Synthetic; good cups, hours

Essential fields: `runId, skuId, cups, productionWindow, releaseDate, status`

CapacityWindow

Key: `windowId` **Grain:** One reserved capacity window **Units / classification:** Synthetic; hours

Essential fields: `windowId, allocatedHours, rate, changeoverRule`

Component

Key: `componentId` **Grain:** One purchasing component **Units / classification:** Synthetic; kg/sets/cartons

Essential fields: `componentId, purchasingUnit, price, orderMultiple, leadTime`

PlanningBOM

Key: `skuId + componentId` **Grain:** SKU-component coefficient **Units / classification:** Synthetic; quantity/good cup

Essential fields: `skuId, componentId, quantityPerCup, planningOnlyFlag`

ComponentStock

Key: `componentId + lotId` **Grain:** Component stock lot **Units / classification:** Synthetic; component unit

Essential fields: `componentId, quantity, availableDate, paidStatus`

ComponentPO

Key: `poId + lineId` **Grain:** One PO component line **Units / classification:** Synthetic; unit and USD

Essential fields: `poId, componentId, quantity, receiptDate, paidAmount, unpaidAmount`

AllocationPolicy

Key: `skuId + nodeId + horizon` **Grain:** One protection rule **Units / classification:** Synthetic; demand/buffer cups

Essential fields: `skuId, nodeId, protectedDemand, buffer, destinationPermission`

TransferOption

Key: `transferId` **Grain:** One transfer candidate **Units / classification:** Synthetic; cups, USD, arrival

Essential fields: `skuId, source, destination, quantity, arrivalDate, expense, eligibility`

CostProfile

Key: `skuId` **Grain:** One SKU cost profile **Units / classification:** Synthetic; USD/good cup

Essential fields: `skuId, materialCost, conversionCost, netRevenueAssumption`

Payment

Key: `paymentId` **Grain:** One dated payment obligation **Units / classification:** Synthetic/derived; USD

Essential fields: `paymentId, commitmentId, dueDate, amount, paidOrFuture`

ForecastLedger

Key: `issue + week + skuId + method` **Grain:** One archived forecast row **Units / classification:** Synthetic forecast / observed fixture

Essential fields: `forecastIssuedAt, targetWeek, skuId, method, forecast, actual`

Scenario

Key: `scenarioId + snapshotId` **Grain:** One scenario preset/override **Units / classification:** Synthetic override, derived outputs

Essential fields: `scenarioId, overrides, optionIds, snapshotId`

Recommendation

Key: `decisionId` **Grain:** One proposed decision **Units / classification:** Analyst interpretation

Essential fields: `decisionId, evidenceRefs, action, quantity, timing, confidence, clientQuestion`

AnalystReview

Key: `reviewId` **Grain:** One illustrative review **Units / classification:** Analyst interpretation

Essential fields: `reviewId, snapshotId, assumptions, interpretation, reviewedAt`

DecisionLog

Key: `decisionId + eventId` **Grain:** One decision-status event **Units / classification:** Synthetic; no client approval

Essential fields: `decisionId, proposedStatus, clientResponse, executionStatus`

DataQualityIssue

Key: `issueId` **Grain:** One raw-input quality issue **Units / classification:** Synthetic negative-test fixture

Essential fields: `issueId, entityId, rule, severity, exclusionStatus`

Referential and type contract

• `skuId` references PlanningSKU; PlanningSKU references Product. Store references Retailer and optional cohort. Every time-bucketed record references an ISO week-start date and the same snapshot.

• ProductionRun references CapacityWindow. PlanningBOM joins PlanningSKU to Component. ComponentPO references Component; Payment references its PO, run or transfer commitment.

• BrandInventoryLot references SKU and node. TransferOption references donor and destination nodes and compatible lots. Recommendation links evidence records, assumptions and alternatives by stable IDs.

• Store-week packs and booked movements are nonnegative integers. Consumer pack quantity converts to cups through explicit metadata. Forecast expected cups may be fractional. Money uses fixed-decimal USD; coefficients retain full precision. Hours and kilograms use numeric types without display-rounding in calculations.

• Date-only values use ISO 8601. Unknown dates, terms, prices, permissions and review identities remain null with a reason. Zero means a measured or supplied zero; it never stands in for missing data.

• Add lineId, transferId, lotId, scopeId, eventId and stable source/assumption IDs where absent. These are technical identifiers only. Never invent client approvals or operational timestamps.

Input and output contract

```
calculate(snapshot, optionId, scenarioOverrides) -> {
  snapshotId, optionId, scenarioId,
  skuWeek: [required, receipts, served, unmet, closing, buffer],
  portfolio: [required, served, unmet, closing],
  capacity: [runHours, changeoverHours, totalHours, feasible],
  components: [componentId, required, availableByRun, shortage],
  allocation: [eligible, donorResidual, arrivalBeforeNeed, expense],
  economics: [materialConsumption, conversionCost, productionCost],
  cash: [datedFuturePayments, previouslyPaid, unresolvedTerms],
  violations: [ruleId, entityId, severity, explanation],
  recommendationStatus: 'exploratory' | 'reviewed-baseline'
}
```

Inputs are immutable. Reject invalid bounds and unmapped units before calculation. Return structured violations instead of silently fixing an infeasible option. A blocked club-pack outcome must remain blocked rather than acquiring invented supply. All pages consume selectors from this output; display-only formatting happens afterward.

Reconciliation pipeline

Validate keys and units -> reconcile store/cohort/SKU panels -> reconcile retail receipts and POS -> calculate demand without duplicate firm orders -> check production/components/release dates -> simulate brand stock -> check node and lot eligibility -> calculate cost and dated cash -> assemble evidence and interpretation. Control-panel rows never enter operating demand. Retail balances never enter brand stock.

06.01 Data brief / Weekly Decision Brief

Required datasets

• Canonical SKU inventory.

• Forward dispatch forecast.

• Requested and reviewed production.

• Bridge commitments.

• Component availability.

• Cash schedule.

• Regional allocation.

• Analyst recommendations.

Four primary decisions

ID	SKU	Action	Quantity/timing implication	Supporting evidence
D01	CC01	REVIEW PRODUCTION	Compare 8,000 with 12,000 next-window cups	Launch volume, portfolio movement, cohort differences, component constraints
D02	CC02	PROTECT CAPACITY	Preserve 12,000 cups for W3	Only two weeks of opening cover; displacement creates timing exposure
D03	CC04	REALLOCATE	Move 1,200 cups before regional depletion	Adequate network stock, West shortfall, East compatible surplus
D04	CC05	HOLD ADDITIONAL DEPTH	Limit next-window run to 2,000	Six weeks of opening cover; 4,000-cup bridge already planned

Secondary dependencies

• CC03 needs 2,000 bridge cups available in W5.

• CC01 and CC02 have little timing resilience before W3.

• CC04 needs another commitment reviewed before stock reaches zero at W8.

Executive constants

Metric	Value
Products modeled	8
Operating retailer panels	3
Immediate primary decisions	4
Opening brand-owned stock	42,000 cups
Current operating-panel POS	13,600 cups/week
Reviewed next-window output	40,000 cups
Reviewed production cost	$43,700
Reviewed first-window future cash including transfer	$18,600
Required next-window line time	36.83 hours

Required analyst narrative

> The launch deserves production, but not an automatic expansion equal to its headline sales. Protect the proven incumbent requirement, resolve the regional imbalance and preserve flexibility in flavors already carrying more cover.

This is synthetic reviewed interpretation.

06.02 Data brief / Launch Incrementality

Required datasets

• Twenty-week Target history.

• Four prelaunch anchor weeks.

• Launch-period SKU history.

• Matched comparison groups.

• Cohort availability and commercial-event records.

Required calculations

```
Observed portfolio growth: 1,100
Incumbent movement: -1,300
Launch sales: 2,400
Descriptive net-growth ratio: 45.8%

Modeled no-launch portfolio: 10,300
Associated difference: 800
Sensitivity range: 300-1,300
```

Decision being prepared

Should the next run add total output, change the mix, or wait for more evidence before expanding?

What ScaleSight contributes

• Separates product success from portfolio requirement.

• Tests alternative explanations.

• Identifies cohorts providing stronger evidence.

• Connects the analysis to an actual production alternative.

Prohibited inference

Do not call the 1,300-cup incumbent decline confirmed cannibalization.

Do not claim the collaboration is weak: it can be commercially valuable even when some sales shift between products.

06.03 Data brief / Reorder Quality

Required datasets

• Initial-fill receipts.

• Consumer POS.

• Subsequent receipt/order cycles.

• Retailer inventory.

• Listed doors.

• Available selling days.

• Ordering-policy assumptions.

Canonical evidence

Cohort	Later receipt cycles	Closing collaboration stock	Planning question
A	3	2,384	Does portfolio movement support more sustained depth?
B	1	136	Is another reorder needed for service, despite flat portfolio growth?
C	0	232	Is weaker movement explained by demand, availability or assortment fit?
D	2	1,228	Does persistence justify a stronger allocation?

Decision being prepared

Which replenishment requirements are supported by consumption - and which production increases still depend on unproven persistence?

Evidence rule

A reorder is an input, not proof.

It must be assessed alongside:

• Consumer sales.

• Stock remaining.

• New doors.

• Available days.

• Order batching.

• Changes in retailer policy.

06.04 Data brief / Store Cohorts

Required datasets

• Store master.

• SKU × store × week POS.

• Availability.

• Price/promotion context.

• Cohort assignments.

• Comparison-store history.

• Counterfactual calculations.

Required cohort outputs

Cohort	Launch sales	Observed portfolio movement	Counterfactual-associated difference
A	900	+800	+700
B	800	0	-50
C	200	0	+50
D	500	+300	+100

Decision being prepared

Should every store group receive the same production-backed allocation?

Analytical contribution

A and B both have strong launch sales but different portfolio patterns.

That gives management a more specific question than whether the collaboration is “doing well.”

Data-science implementation boundary

The supplied cohorts are defined fixtures.

If Wahib later implements fitted clustering, retain:

```
clusterMethod
featureSet
trainingWindow
fitVersion
assignmentConfidenceOrDistance
```

Do not relabel manually supplied cohorts as ML output.

06.05 Data brief / Production Priorities

Required datasets

- Requested/reviewed quantities.
- Run rates.
- Changeovers.
- Allocated line hours.
- Component stocks.
- Release dates.
- Inventory trajectories.
- Bridge commitments.

Required comparison

Measure	Requested	Reviewed
Cups	52,000	40,000
Hours	46.83	36.83
Generic packaging requirement	52,000	40,000
Base requirement	4,160 kg	3,200 kg
Production-cost exposure	$56,900	$43,700

Decision being prepared

Which commitments justify the allocated window, and what would be displaced by expanding the collaboration?

Important interpretation

The reviewed mix increases Cookies relative to the request and limits several flavors with more stock.

It is an **analyst-supplied feasible alternative**, not a claimed optimized mathematical solution.

The engine must explain feasibility and commercial consequences, not simply announce “optimal.”

06.06 Data brief / Components & Commitments

Required datasets

- Public component categories.
- Synthetic planning BOM.
- Component stocks.
- Printed carton stock.
- Component POs.
- Supplier timing and terms.
- Production alternatives.

Required worked example

For CC01:

```
8,000 cups → 96 kg granola
10,000 cups → 120 kg granola
12,000 cups → 144 kg granola
```

At 12,000 cups, current granola availability is insufficient before the production window.

Alternatives

Alternative	Consequence
Hold at 8,000	24 kg granola remains before later receipts
Increase to 10,000	Uses all current granola and strawberry shell allocated to that SKU
Increase to 12,000	Requires additional components and wider capacity/cash changes
Reserve another 60 kg	Creates $300 commitment; does not itself solve production capacity

Decision being prepared

Should Alec’s commit finished output now - or preserve a component position while evidence develops?

Reservation terms must remain visible as incomplete.

06.07 Data brief / Allocation

Required datasets

• Brand-owned stock by location.

• Region-level dispatch requirements.

• Protected requirements.

• Lot eligibility.

• Transfer lanes.

• Transit and release times.

• Transfer cost.

Canonical decision

Move **1,200 CC04 cups** from East to West.

Results:

• West’s two-week modeled gap falls from 1,200 to zero.

• East retains protected demand and buffer.

• East’s remaining transferable surplus is 720.

• Network stock remains 6,000.

• Incremental expense is $400.

Decision being prepared

Is an additional run necessary, or can eligible existing inventory arrive before the requirement?

Mandatory distinction

Changing future flavor output is production-mix allocation.

Moving existing CC04 stock is finished-goods reallocation.

They are different actions with different constraints.

06.08 Data brief / Cash Exposure

Required datasets

• SKU material/conversion costs.

• Material stock already paid for.

• Unpaid PO balances.

• Conversion payment dates.

• Transfer expense.

• Alternative purchases.

• Finished-stock projections.

• Synthetic net revenue assumptions.

Required outputs

• Production cost consumed.

• Incremental future cash.

• Previously paid amounts.

• Future committed payments.

• Uncommitted optional purchases.

• Finished-stock exposure.

• Modeled unserved-dispatch exposure.

Canonical comparisons

```
Requested production cost: $56,900
Reviewed production cost: $43,700

Reviewed first-window future cash:
$18,200 conversion + $400 transfer = $18,600

Eight-week future cash:
$52,265
```

Decision being prepared

Which option protects requirements while committing an acceptable amount of new cash?

Important correction

For an in-house manufacturer, unit production cost is not automatically the amount of cash that must be paid tomorrow.

The model must show that distinction.

06.09 Data brief / Forecast Learning

Required datasets

• Historical POS and availability.

• Archived synthetic forecast ledger.

• Forecast issue dates.

• Actual observation dates.

• Launch-age information.

• Promotion and price flags.

• Method/version metadata.

• Analyst overrides and reasons.

Required calculations

From the supplied forecast ledger:

• Naive WAPE: 5.46%.

• Reviewed illustrative WAPE: 3.17%.

• Naive bias: +3.71%.

• Reviewed illustrative bias: +1.33%.

Decision being prepared

Which changing signals should alter the next planning assumption?

What the page should establish analytically

• A model must be compared with a baseline.

• Forecast error and business decision quality are related but different.

• Sparse launch history needs pooling and judgment.

• A forecast improvement is useful only if it changes a feasible commitment.

Honest capability statement

Statistical and ML methods can support the investigation.

The current synthetic forecast ledger is not proof that a real Alec’s ML model has been trained or validated.

06.10 Data brief / Scenario Lab

Required datasets

- Canonical snapshot.
- Scenario input changes.
- Production alternatives.
- Component constraints.
- Inventory trajectories.
- Cash schedules.
- Allocation eligibility.
- Analyst interpretation.

Required outputs per scenario

```
required dispatch
served dispatch
unmet dispatch
closing stock
first shortage period
first buffer breach
production hours
component shortages
future cash
incompatible actions
review question
```

Decision being prepared

> What would change the recommendation - and which response addresses the actual cause?

Examples:

- Stronger demand before W3 requires timing intervention.
- Earlier orders may change timing without changing total demand.
- More line time may not help if components are unavailable.
- Club demand cannot be served from an incompatible pack.
- Delayed output can leave more ending stock while missing earlier requirements.

The scenario engine should make those relationships explicit.

06.11 Data brief / Managed Planning

Required service dataset

Use a synthetic review record for each primary decision.

```
decisionId
snapshotId
skuId
signal
investigation
optionsCompared
recommendedAction
quantity
requiredTiming
productionCostEffect
futureCashEffect
evidenceConfidence
executionConfidence
assumptions
clientQuestion
nextReviewDate
status
```

Example review record

```
Decision: D01
Product: CC01

Signal:
Strong launch sales alongside weaker incumbent movement.

Investigation:
Compare portfolio movement, control-panel behavior,
availability, receipt cycles and production feasibility.

Options:
12,000 requested cups
8,000 reviewed cups
component reservation
later expansion after another evidence cycle

Recommendation:
Commit the feasible partial run;
retain a defined review of additional depth.

Client question:
Are there retailer commitments or supplier constraints
not represented in these assumptions?

Status:
Illustrative proposal  -  client decision not recorded.
```

Ownership

Work	Owner in proposed service
Existing operational data and business context	Alec’s
Forecast maintenance and investigation	ScaleSight
Supply-chain feasibility review	ScaleSight with Alec’s team
Prepared options and decision brief	ScaleSight
Final quantity, policy and commercial decision	Alec’s
Execution through existing systems	Alec’s
Review of what happened afterward	Joint

Pilot boundary

A real pilot could focus on:

> One upcoming Culture Cup production window, its retailer requirements, components and cash implications.

The demo remains broad enough to show the full capability.

07.01 Phase 2 / Brand direction

Make it feel like Alec’s

Use a warm, product-led visual identity, with the Culture Cup products visible throughout the experience.

The analytical pages should feel commercially serious while retaining the character of an ice-cream brand.

Proposed palette - not verified official brand tokens:

Use	Color
Main background	Warm cream #FFF8EE
Primary text	Deep green #21372F
Cards	Soft white #FFFEFA
Borders	Warm neutral #E5DFD4
Secondary text	Muted green #66736A
Collaboration accent	Strawberry pink #E98AA3
Highlight	Butter yellow #F1D46A
Supporting accent	Sage #8FB69C
Coffee accent	Coffee brown #997357
ScaleSight attribution	Navy #172B45

Typography

• Expressive, rounded headings.

• Highly readable body and table text.

• Tabular numbers for quantities, money and percentages.

• Use licensed fonts; do not assume access to Alec’s exact brand font.

Product imagery

Use genuine public product images for the eight Culture Cup products.

Normalize image size and positioning. Preserve packaging proportions and use optimized local assets.

Product images should help users recognize the commitment being discussed. They should not displace useful analysis.

Visual variety

Each page needs its own main visual:

• Decision cards.

• Portfolio waterfall.

• Receipt and sales timeline.

- Cohort scatterplot.
- Production schedule.
- Component dependency diagram.
- Regional allocation comparison.
- Cash timeline.
- Forecast comparison.
- Scenario projection.
- Worked service example.

Avoid building 11 versions of the same KPI-card dashboard.

Status and product treatment

Status colors remain separate from flavor colors. Retain original packaging proportions and colors. Do not fabricate packaging, stretch cartons or use generated substitutes. Product images are implementation asset requirements; the PDF uses a text-led brand treatment.

07.02 Phase 2 / Navigation & shared state

Workspace title

Alec’s × ScaleSight

Secondary title:

Retail-to-Production Planning

Supporting description:

Retail signals, reviewed against production, components and cash.

Navigation

Group	Page	Route
Weekly review	Weekly Decision Brief	`/`
Demand evidence	Launch Incrementality	`/incrementality`
Demand evidence	Reorder Quality	`/reorder-quality`
Demand evidence	Store Cohorts	`/cohorts`
Production commitments	Production Priorities	`/production`
Production commitments	Components & Commitments	`/components`
Production commitments	Allocation	`/allocation`
Production commitments	Cash Exposure	`/cash`
Analytical review	Forecast Learning	`/forecast-learning`
Analytical review	Scenario Lab	`/scenarios`
ScaleSight service	Managed Planning	`/managed-intelligence`

The evidence drawer is shared across these pages. It is **not a twelfth page**.

Persistent header

Show:

• Alec’s × ScaleSight.

• Current page.

• **Illustrative data** badge.

• Review date: **12 October 2026**.

• Current state: **Reviewed baseline** or the selected exploration.

• A visible **Reset exploration** control when a scenario is active.

Persistent disclosure

> Illustrative planning concept. Operating data and recommendations are synthetic. No live connection or client results.

Footer:

> Analysis maintained by ScaleSight. Decisions made with the client.

Keep disclosure readable without making it the dominant visual.

Navigation behavior

• Every route opens directly.

• Product context persists when following a related decision.

• Scenario context persists between analytical pages.

• The reviewed baseline is always recoverable.

• Browser back/forward navigation works.

• Returning to a page does not reset the current investigation unexpectedly.

Shared interaction and evidence system

Three separate states

State	Meaning
Source fixtures	Immutable Phase 1 synthetic inputs
Reviewed baseline	The illustrative comparison and recommendations prepared for the review
Exploration	A user-selected scenario calculated from those inputs

Changing a scenario must not silently rewrite the reviewed recommendation.

An active exploration should show:

> **Exploring: Collaboration demand +25%**

And:

> Simulated outcome. A changed recommendation requires specialist review and client context.

Shared evidence drawer

Every important result should offer **View evidence**.

The drawer contains:

1. Business question.

2. Product, retailer and period covered.

3. Input values.

4. Evidence classification.

5. Calculation or comparison.

6. Assumptions.

7. Relevant constraints.

8. What remains unknown.

Use the four Phase 1 classifications consistently:

• **Public fact**

• **Synthetic input**

• **Derived metric**

• **Analyst interpretation**

Valid interactions

Users can:

• Select a product or cohort.

• Compare options.

• Inspect assumptions.

• Explore bounded scenarios.

• Examine inventory and cash consequences.

• Reset to baseline.

• Follow a recommendation into its supporting analysis.

The demo should not contain operational execution controls such as approving production, sending a PO or executing a transfer.

Page anatomy and analyst treatment

Each page should contain these elements, adapted to its purpose:

1. **A founder-readable question.**

2. **A short explanation of why it matters.**

3. **The main analytical comparison.**

4. **A concise ScaleSight interpretation.**

5. **The commitment or decision being considered.**

6. **An evidence/calculation link.**

7. **A relevant next-page link.**

Avoid filling every page with the same five-card template.

Analyst interpretation format

Use a compact block:

ScaleSight review

Two or three sentences explaining:

• What the evidence suggests.

• What it does not establish.

• What decision follows.

Then:

Decision for Alec’s

One concrete question.

Evidence drawer

Opening **View evidence & assumptions** should show:

Item	Content
Scope	Target panel, operating panel, factory window or specific location
Inputs	Relevant records
Classification	Public / Synthetic / Derived / Interpretation
Calculation	Formula and worked values
Assumptions	The conditions supporting the result
Limitation	What cannot be concluded
Source	Public source when applicable

The drawer must show the relevant evidence for the selected result, not the same generic disclaimer everywhere.

Bounded controls

Adopt the following supplied Phase 2 control limits (technical UI limits, not validated probabilistic ranges):

Input	Allowed range
Demand change	-30% to +40%
Timing shift	-2 to +4 weeks
Component availability	0% to 120% of baseline

Invalid input:

Enter a value within the supported scenario range.

Infeasible output:

This option exceeds the available constraint. Review the highlighted requirement before treating it as executable.

Do not silently cap or modify the visitor’s input.

Page specifications

07.03 Page 1 / Weekly Decision Brief

Route: /

Purpose

Give the founder the commercial story immediately: four commitments deserve attention, and ScaleSight has worked through the evidence behind them.

Exact opening copy

Eyebrow

WEEKLY PLANNING REVIEW

Headline

> The fastest-moving SKU isn’t automatically the SKU that deserves more production.

Supporting line

> Four commitments deserve a closer look before the next production window.

Main visual: four product decision cards

Card 1 - Collaboration Culture Cup

Label: REVIEW PRODUCTION

Headline

> Strong launch sales. A different portfolio story.

Show:

• Latest weekly collaboration sales: **2,400 cups**.

• Observed Target portfolio growth: **1,100 cups**.

• Requested next run: **12,000 cups**.

• Reviewed candidate: **8,000 cups**.

Short interpretation:

> Launch sales are substantial, but the wider portfolio movement gives us a different question about how much additional production to commit.

Action:

Examine the launch signal →

Links to `/incrementality`.

Card 2 - Cookies & Cream

Label: PROTECT CAPACITY

Headline

> Don’t let the launch displace the next requirement.

Show:

• Opening brand stock: **6,000 cups**.

• Current weekly demand: **3,000 cups**.

• Next reviewed release: **Week 3**.

• Reviewed next run: **12,000 cups**.

Interpretation:

> The existing stock covers the first two weeks exactly. Production timing matters alongside the quantity.

Action:

Review the production slot →

Links to `/production`, with CC02 selected.

Card 3 - Chocolate Covered Strawberry

Label: REALLOCATE

Headline

> Enough stock overall. A regional shortage underneath.

Show:

• West requirement gap: **1,200 cups**.

• Proposed East-to-West transfer: **1,200 cups**.

• Transfer and handling expense: **$400**.

Interpretation:

> Compatible stock elsewhere could resolve the regional gap before another run is needed.

Action:

Compare the allocation →

Links to `/allocation`.

Card 4 - Mint Chocolate Cookie

Label: HOLD ADDITIONAL DEPTH

Headline

> More stock does not yet require a deeper run.

Show:

• Opening stock: **7,200 cups**.

• Current cover: **6.0 weeks**.

• Requested run: **4,000 cups**.

• Reviewed candidate: **2,000 cups**.

Interpretation:

> Existing stock and the later bridge receipt reduce the case for a larger immediate commitment.

Action:

Inspect the buying depth →

Links to `/production`, with CC05 selected.

Production-window summary

Below the cards:

Measure	Reviewed candidate
Output	40,000 cups
Line time, including changeovers	36.83 hours
Modeled production cost	$43,700
Future cash for this window, including transfer	$18,600

Label this explicitly as the **next production window**, rather than weekly business totals.

Closing prompt

> What changes if demand strengthens - or the release arrives a week late?

Link to Scenario Lab.

Additional dependency note

> **Timing still matters:** the reviewed plan depends on the W3 release and the early Honeycomb bridge tranche.

Link to `/scenarios`.

Primary walkthrough action

> **Walk through the launch decision**

Starts at Incrementality.

Acceptance

Four primary decisions appear.

- Product images correspond to the correct SKU.
- All figures come from shared selectors.
- Cash and production cost remain separate.
- No implied client approval.
- Under a scenario, reviewed cards remain identifiable as baseline interpretation.

07.04 Page 2 / Launch Incrementality

Route: `/incrementality`

Business question

Does strong collaboration sales justify more total Culture Cup production, or a change in the flavor mix?

Exact copy

Headline

Is the collaboration growing Culture Cup - or changing the flavor mix?

Supporting line

Strong launch sales can lead to different production decisions depending on what happens across the portfolio.

Main visual: observed portfolio waterfall

Show:

Movement	Cups per week
Pre-launch Target portfolio	10,000
Collaboration sales	+2,400
Incumbent product movement	-1,300
Latest Target portfolio	11,100

Use **Incumbent product movement**, rather than “proven cannibalization.”

Below:

Collaboration sales: **2,400 cups** Observed net portfolio growth: **1,100 cups**

Two analysis views

View A - Observed movement

Default view.

Show the before-and-after SKU mix, launch trajectory and availability context.

View B - Counterfactual estimate

Show:

• Estimated portfolio without launch: **10,300 cups**.

• Observed portfolio: **11,100 cups**.

• Associated difference: **800 cups**.

• Assumption sensitivity: **300-1,300 cups**.

Required explanation:

> This comparison uses synthetic matched controls. It estimates an associated difference under stated assumptions; it does not establish customer-level causality.

Do not present the sensitivity range as a statistical confidence interval.

Reviewed interpretation

> The launch warrants continued attention. The portfolio evidence supports reviewing the production mix before treating all launch sales as additional demand.

Client decision

> Should the next commitment increase total output, change the flavor mix, or wait for another cycle of evidence?

Interaction

- Toggle observed/counterfactual.
- Select a SKU.
- Inspect cohort contributions.
- Open assumptions.
- Continue to Reorder Quality.

Interactions

Observed movement / Counterfactual estimate

This switches the analytical explanation, not the underlying data.

View by cohort

Opens the cohort breakdown or navigates to `/cohorts`.

What could explain the difference?

Evidence drawer sections:

- Availability.
- Price and promotions.
- Distribution.
- Comparison-store behavior.
- Unobserved customer switching.

Next link

> **Check whether retailer consumption supports the next run →**

Destination: `/reorder-quality`.

Acceptance

Both analytical layers are available.

- The 45.8% ratio, if displayed, says **Observed net-growth ratio**.
- The 33.3% ratio, if displayed, says **Counterfactual-associated lift ratio**.

- Neither is called true incrementality.
- Target-panel scope remains visible.
- No customer-level causal claim.

07.05 Page 3 / Reorder Quality

Route: `/reorder-quality`

Exact copy

Headline

When has a launch earned a deeper commitment?

Supporting line

Opening fill, consumer sales and later replenishment each tell us something different.

Main visual: receipt-to-consumption timeline

For each cohort, show three aligned lanes:

1. Retailer receipts.

2. Consumer POS.

3. Closing retailer-owned inventory.

Periods:

• 14 September.

• 21 September.

• 28 September.

• 5 October.

Use clear visual distinctions between a shipment into retail and a sale to a consumer.

Cohort selection

Cohort	Later receipt cycles	Closing retailer stock	Interpretation to investigate
A	3	2,384 cups	Stronger portfolio movement
B	1	136 cups	Replenishment need despite flat portfolio
C	0	232 cups	Availability and assortment context
D	2	1,228 cups	More persistent later-cycle movement

Important analytical distinction

For Cohort B:

Low remaining stock can justify service replenishment even when the launch has not increased total portfolio sales.

This makes the page useful beyond a simplistic “reorders prove success” story.

Evidence checklist

Show categorical states, with explanations:

• Receipts reconcile to inventory.

• Consumer POS is available.

• Availability has been considered.

• Distribution changes are understood.

• Promotion context is known or missing.

• Portfolio movement has been examined.

Do not invent a numerical reorder-quality score.

Client decision

Which receipts protect existing demand, and which signals justify a larger future commitment?

Cohort selector

A · B · C · D

Default: A.

Each selection updates:

• Receipt events.

• Weekly POS.

• Cumulative receipts.

• Closing stock.

• Available selling exposure.

• Interpretation.

Cohort-specific interpretation

A

Later receipt cycles and stronger portfolio movement provide more evidence to investigate sustained depth.

B

Strong launch consumption has left little retailer stock. Replenishment may be needed even though total portfolio demand is flat.

C

Weaker launch movement needs an availability and assortment check before reducing the commitment.

D

Continued consumption through later receipt cycles supports another review of allocation and run depth.

Next link

Compare the store groups →

Destination: `/cohorts`.

Acceptance

Receipts and POS cannot be mistaken for one another.

- Inventory arithmetic reconciles.
- Retailer-owned inventory is labeled.
- Reorder count is never a standalone decision rule.
- No repeat-purchase probability is invented.

07.06 Page 4 / Store Cohorts

Route: `/cohorts`

Exact copy

Headline

> The same launch sales can mean different things in different stores.

Supporting line

> Compare launch velocity with availability and movement across the rest of the portfolio.

Main visual: cohort scatterplot

- **X-axis:** Collaboration cups per available store-week.
- **Y-axis:** Observed portfolio change versus the pre-launch period.
- Four labeled points: A, B, C and D.

Use a consistent cohort color system across this page and Reorder Quality.

Supporting cohort comparison

Cohort	Latest portfolio	Pre-launch portfolio	Collaboration availability
A	3,300 cups	2,500 cups	95.2%
B	2,500 cups	2,500 cups	93.3%
C	2,500 cups	2,500 cups	80.0%
D	2,800 cups	2,500 cups	97.1%

Selecting a cohort opens:

- 30-store scope.
- Launch sales.
- Availability.
- Before-and-after product mix.
- Matched-control comparison.
- Receipt history.
- Interpretation and uncertainty.

Method disclosure

> These cohorts are defined in the synthetic dataset. Automated clustering could be evaluated with real data during an engagement.

Do not imply a clustering model was trained.

Reviewed interpretation

Strong launch velocity and stronger portfolio demand are different signals. Availability also changes how confidently we can interpret a weaker result.

Client decision

Where does the evidence support additional depth, and where should replenishment preserve the existing mix?

Investigation labels

Use:

- **Stronger portfolio movement**
- **High launch, flat portfolio**
- **Incumbent-led mix**
- **Persistent launch consumption**

These are descriptive, not causal conclusions.

Interactions

Select cohort.

- Compare two cohorts.
- Toggle observed versus counterfactual portfolio difference.
- Inspect availability.
- Open the relevant SKU history.

The selected comparison must update all associated values.

Next link

Translate the evidence into production →

Destination: `/production`.

Acceptance

Cohorts reconcile to the Target total.

- Availability is included.
- Counterfactual differences retain their qualifications.
- Defined cohorts are not mislabeled as trained ML clusters.
- No real store identities are implied.

07.07 Page 5 / Production Priorities

Route: `/production`

Exact copy

Headline

Where should the next production slot go?

Supporting line

Compare the requested mix with a feasible candidate across demand, line time, components and upcoming supply.

Main visual: production schedule comparison

Show two schedules:

- **Requested mix**
- **Reviewed candidate**

Run widths represent line hours. Include changeover blocks.

Constraint	Requested	Reviewed	Available
Output	52,000 cups	40,000 cups	Component-dependent
Line time	46.83 h	36.83 h	40.00 h
Cup-and-lid sets	52,000	40,000	40,000
Base	4,160 kg	3,200 kg	3,200 kg

Mark the requested mix **Infeasible under current assumptions**.

Mark the reviewed mix **Feasible candidate for discussion**.

SKU comparison table

Product	Requested	Reviewed
Collaboration	12,000	8,000
Cookies & Cream	10,000	12,000
Dark Chocolate Honeycomb Crunch	6,000	4,000
Chocolate Covered Strawberry	6,000	6,000
Mint Chocolate Cookie	4,000	2,000
GROUNDWORK Coffee	4,000	2,000
Madagascar Vanilla Bean	6,000	4,000
Peanut Butter Cup	4,000	2,000
Total	**52,000**	**40,000**

Selected-product panel

Show:

- Opening stock.
- Forward demand.
- Requested and reviewed quantities.
- Release week.
- Confirmed later receipts.
- Weekly inventory trajectory.
- Binding components.
- Why the quantity merits review.

Critical note

> The reviewed mix leaves 3.17 line hours available, but consumes the available base and cup-and-lid sets.

Unused line time does not mean more production is feasible.

Also flag:

> Dark Chocolate Honeycomb Crunch depends on its earlier Week 5 bridge receipt.

Interactions

- Select a product.
- Toggle requested/reviewed.
- Inspect a run’s dependencies.
- Follow a component into Components.
- Follow cost into Cash.

• Test release timing in Scenario Lab.

Product selection

Selecting a product updates:

• Requested output.

• Reviewed output.

• Opening cover.

• Upcoming receipts.

• Projection through W8.

• Component requirement.

• Analyst explanation.

Important product explanations

CC01

The reviewed run retains meaningful launch output while limiting the larger commitment that requires additional evidence and components.

CC02

Preserve output for the established dispatch requirement. Opening stock has little resilience before the W3 receipt.

CC05

Existing stock and the bridge commitment support less additional depth in this window.

CC03

The reviewed quantity depends on the early W5 bridge tranche. Keep that timing visible.

Decision for Alec’s

Which output should be committed in this window, and what would need to change to justify the larger request?

Acceptance

Zero-quantity runs do not create changeovers.

• Minimums and increments are enforced.

• Chart widths represent actual run hours.

• Production quantity and weekly demand are not conflated.

• Component and timing constraints remain visible.

07.08 Page 6 / Components & Commitments

Route: `/components`

Exact copy

Headline

A deeper run creates more than finished inventory.

Supporting line

See what the next quantity requires - and which commitments can remain flexible.

Main visual: selected-product dependency diagram

For the collaboration product:

Proposed run →

• Ice-cream base.

• Strawberry shell.

• Granola.

• Cup-and-lid sets.

• Printed cartons.

• Production time.

Keep ingredient quantities labeled as **synthetic planning coefficients**, rather than actual recipes.

Collaboration quantity comparison

Run	Granola	Strawberry shell
8,000 cups	96 kg	120 kg
10,000 cups	120 kg	150 kg
12,000 cups	144 kg	180 kg
Available	120 kg	150 kg

Highlight the shortages under the 12,000-cup option.

Optional commitment card

Optional granola purchase

• Quantity: **60 kg**.

• Value: **$300**.

• Deposit: **$90**.

• Later balance: **$210**.

• Earliest standard arrival: **26 October**.

• Cancellation terms: **Unknown**.

Interpretation:

> Reserving an ingredient may preserve an option. It does not reserve production capacity or authorize finished-goods output.

Additional interactions

For every SKU:

• View component coverage.

• Inspect incoming component POs.

• Compare existing stock with required consumption.

• Show the binding constraint.

• Inspect supplier timing assumptions.

Client decision

> Which commitments should be made now, which can preserve flexibility, and which need stronger demand evidence?

Interpretation by alternative

8,000

> Fits the selected component position.

10,000

> Uses all currently available granola and strawberry shell for this SKU.

12,000

> Requires additional components before the window, alongside wider production changes.

Public/synthetic distinction

Near the BOM:

> **Public ingredient categories. Synthetic planning coefficients.**

Keep the fuller qualification in the evidence drawer.

Interactions

Select any of the eight products.

• Compare predefined production quantities.

• Inspect available versus required component stock.

• Trace incoming PO.

• Compare reserve/produce/wait options.

Acceptance

All eight products have working dependencies.

- CC01 does not share every inclusion with incumbents.
- Shared components reconcile across products.
- Existing POs and optional purchases remain distinct.
- No implied real recipe or supplier commitment.

07.09 Page 7 / Allocation

Route: `/allocation`

Exact copy

Headline

Do we need another run - or a better allocation?

Supporting line

Test whether compatible stock can resolve the shortage before committing more production.

Main visual: regional stock comparison

Use three regional panels with a proposed transfer between East and West.

Before transfer

Region	Stock	Two-week demand	Protected buffer	Position
West	600	1,800	0	1,200-cup gap
East	3,600	1,000	680	Transferable stock
Central	1,800	1,200	-	No transfer proposed

Proposed transfer

1,200 cups / 50 cases: East → West

- Transit: **2 days**.
- Receiving: **1 day**.
- Freight, handling and QA: **$400**.

After transfer

- West requirement gap: **0 cups**.
- East remaining stock after transfer: **2,400 cups**.
- East residual after demand and protected buffer: **720 cups**.
- Total network stock: **6,000 cups**, unchanged.

Feasibility checklist

- Same SKU.
- Compatible pack.
- Destination permitted.
- Brand-owned inventory.
- Cold-chain conditions.
- Arrival before projected depletion.

• Donor demand and buffer protected.

• Lot eligibility checked.

Client decision

> Can this transfer arrive in time without creating a shortage elsewhere?

The interaction is a comparison, with no transfer-execution button.

Interactions

Compare before/after.

• Inspect donor protection.

• Inspect transfer eligibility.

• Select another product to see whether a transfer is applicable.

• Examine the hypothetical club restriction.

When an action is incompatible:

> **This stock cannot serve that requirement under the current pack or destination assumptions.**

Next link

> **See the cash and service trade-off →**

Destination: `/cash`.

Acceptance

Network stock is conserved.

• The transfer does not move retailer-owned store stock.

• East protection is calculated, not implied.

• No cross-flavor conversion.

• Club and Target-exclusivity restrictions work.

07.10 Page 8 / Cash Exposure

Route: /cash

Exact copy

Headline

What does this decision commit - and when does cash leave?

Supporting line

Separate the cost of production from the payments still ahead.

Main visual: dated payment timeline

Group payments into:

• Reviewed production conversion.

• Transfer.

• Component PO balances.

• Bridge production conversion.

Show:

Date	Payment	Amount
23 October	Reviewed conversion, first payment	$10,920
26 October	Transfer	$400
30 October	Reviewed conversion, balance	$7,280
2 November	Component PO balances	$16,065
13 November	Bridge conversion, first payment	$10,560
20 November	Bridge conversion, balance	$7,040
Total	**Future payments over eight weeks**	**$52,265**

Cost-versus-cash comparison

Measure	Requested	Reviewed
Modeled production cost	$56,900	$43,700
Future cash for the first window	At least $29,965*	$18,600
First-window cash limit	$25,000	$25,000

*Requested option remains infeasible and excludes recovery costs such as expediting or overtime.

Required interpretation

> The reviewed mix has $13,200 lower modeled production-cost exposure. That is not the same as $13,200 in cash savings.

Show:

• Reviewed first-window cash headroom: **$6,400**.

• Previously paid component deposits: **$6,885**, excluded from future payments.

Interactions

• Toggle **Production cost / Future cash**.

• Expand individual payments.

• Trace a payment to its production run or PO.

• Compare supported scenario consequences.

Client decision

> Which feasible option uses new cash in a way the demand evidence supports?

Inventory exposure

Provide SKU-level closing stock and cover alongside the payment comparison.

Avoid celebrating lower cash if it creates an unacceptable service shortfall.

Interactions

Switch cost/cash view.

• Expand payment group.

• Inspect paid versus future amounts.

• Select scenario.

• Trace a figure to the production or PO record.

Acceptance

No material-cost/PO-payment double count.

• Old deposits are excluded from future cash.

• Net revenue assumptions are not shelf prices.

• No claimed actual savings, margin gain or guaranteed revenue protected.

• Every cash amount has a date and commitment reference.

07.11 Page 9 / Forecast Learning

Route: `/forecast-learning`

Exact copy

Headline

Which signals should change the next planning assumption?

Supporting line

Review forecast differences alongside the demand signals that could change a commitment.

Main visual: product forecast comparison

For the selected SKU, plot:

• Actual synthetic POS.

• Naive forecast.

• Illustrative reviewed forecast.

Use the archived forecast ledger from Phase 1. Forecasts must retain their issue dates.

Portfolio benchmark panel

Metric	Naive	Illustrative reviewed
WAPE	5.46%	3.17%
Signed bias	3.71%	1.33%

Prominent caption:

Calculated from the synthetic demonstration ledger. These are not measured ScaleSight client results.

Analytical questions

Let users explore:

• Is availability suppressing observed sales?

• Is launch demand settling?

• Is the portfolio growing or shifting?

• Are later receipts consistent with consumption?

• Has supply timing changed the relevant planning horizon?

Methods panel

Separate clearly:

Method	Demo status
Availability normalization	Calculated
Forecast error and bias	Calculated

Method	Demo status
Counterfactual comparison	Illustrative method
Forecast paths	Synthetic archived inputs
Hierarchical ML forecasting	Potential engagement method
Automated store clustering	Potential engagement method

Do not show every reviewed forecast as superior. Vanilla should demonstrate that a simple baseline can remain appropriate.

Client decision

Is the difference material enough to change quantities, timing or the next investigation?

Product selection

All eight products must work.

Selecting a product changes:

• Its actual/forecast history.

• Error values.

• Evidence interpretation.

• Related planning implication.

Include examples where the naive method performs well, such as Vanilla.

That makes the analysis more credible than showing universal model superiority.

Signal explanation

Use concise expandable topics:

Availability

Low sales can reflect limited selling opportunity.

Launch age

Early consumption may not persist at the same rate.

Portfolio movement

New-SKU growth may accompany changes elsewhere.

Retailer behavior

Receipt timing can differ from consumer demand.

Supply timing

A better forecast still needs a feasible response.

Method status

Capability	Status wording
Availability normalization	Implemented demo calculation
Forecast-error comparison	Implemented demo calculation
Counterfactual comparison	Illustrative analytical method
Supplied forecast paths	Synthetic archived inputs
Hierarchical ML/clustering	Proposed method unless actually implemented

Acceptance

No fake training animation.

• No fabricated feature importance.

• No confidence percentage.

• No future-data leakage.

• The implemented/proposed distinction is explicit.

• Metrics calculate from records.

07.12 Page 10 / Scenario Lab

Route: `/scenarios`

Exact copy

Headline

What would change the recommendation?

Supporting line

Explore demand and supply changes before a commitment is locked.

Eight selectable options

Baseline plus seven stress scenarios:

Option	Change
Reviewed baseline	Phase 1 reviewed mix
Collaboration softens	Collaboration demand -20%
Collaboration strengthens	Collaboration demand +25%
Cookies & Cream accelerates	Demand +20%
Target requires stock earlier	Collaboration dispatch shifts earlier
Base availability tightens	Available base -20%
Hypothetical club order	New 8-cup pack requirement
Release arrives one week late	Reviewed receipt moves from Week 3 to Week 4

Main visual

A weekly stock trajectory for the selected product, including:

• Required dispatch.

• Released receipts.

• Closing physical stock.

• Buffer.

• Unserved requirement.

Physical stock never becomes negative. Unserved requirement is displayed separately.

Canonical results

Scenario	Eight-week unserved requirement	Ending inventory
Baseline	0	16,800
Collaboration -20%	0	19,920
Collaboration +25%	2,700	15,600
Cookies & Cream +20%	1,200	13,200
Earlier Target dispatch	0	16,800
One-week release delay	6,300	23,100

For the component-constrained candidate:

- Feasible output: **32,000 cups**.
- Modeled unserved requirement: **4,000 cups**.

For the hypothetical club request:

> Blocked pending pack compatibility, permissions and supply review.

Do not suggest an actual Costco order or existing listing.

Important timing interpretation

> More ending stock does not undo demand that could not be served earlier.

The release-delay scenario should make this visually obvious.

Controls

- Product.
- Scenario preset.
- Bounded demand adjustment.
- Receipt timing.
- Component availability.
- Reset baseline.

Use the bounded control ranges in Section 07.02.

Recommendation state

> Candidate for investigation. Specialist review and client context are required before changing the commitment.

Scenario interaction rules

One selected preset supplies the starting assumptions.

- Bounded controls can modify that preset.
- All linked pages use the same exploratory state.

• Every change remains labeled exploratory.

• Reset restores the canonical reviewed baseline exactly.

Scenario-to-page links

Identified issue	Link
Run timing/capacity	Production
Component shortage	Components
Compatible stock elsewhere	Allocation
Payment impact	Cash
Weak evidence	Cohorts / Reorder Quality

Acceptance

Every preset works.

• Results propagate to relevant pages.

• Reset works.

• Infeasible scenarios remain infeasible.

• No hidden capacity increase.

• Stock timing and total quantity remain distinct.

• Club compatibility is enforced.

07.13 Page 11 / Managed Planning

Route: `/managed-intelligence`

Exact copy

Headline

The analysis stays current. Your team keeps the decision.

Supporting line

ScaleSight maintains the forecasts, investigates the exceptions and prepares the comparisons around your next commitment.

Main visual: worked collaboration example

Show the complete service through one decision.

1. Receive the operating picture

Sales, stock, incoming supply, production timing, component availability and commercial plans.

2. Investigate the signal

The collaboration sells strongly, while incumbent sales and store cohorts move differently.

3. Compare feasible options

Requested 12,000-cup collaboration run versus an 8,000-cup candidate within the broader production mix.

4. Review the consequences

Supply-chain review checks service exposure, components, line time, timing and cash.

5. Bring the decision to Alec’s

A short recommendation covering:

• Proposed quantities.

• Timing.

• Evidence.

• Assumptions.

• Cash implications.

• What still requires client judgment.

6. Review the next cycle

Refresh the evidence and investigate whether a change is warranted.

Ownership panel

ScaleSight maintains	Alec’s contributes and controls
Forecasts and analytical comparisons	Commercial plans and operating context
Demand investigations	Final quantities and commitments
Scenario analysis	Supplier and retailer relationships
Reviewed recommendations	Execution through existing systems
Recurring decision brief	Final commercial judgment

Pilot section

Headline

Start with one commitment.

Copy

A focused 30-day planning pilot could examine one upcoming Culture Cup production window using agreed real inputs. We would maintain the comparison and bring your team a reviewed recommendation with quantities, timing, assumptions and cash implications.

CTA

Discuss a production commitment

If no real booking destination is supplied, open a concise explanatory panel. Do not create a fake booking flow.

Prepared recommendation example

Breakfast collaboration

Compare an 8,000-cup commitment with the larger request. The reviewed quantity fits the current window while retaining an evidence review before deeper expansion.

Evidence

Launch consumption, portfolio movement, cohort differences and component availability.

Client context needed

Retailer commitments, supplier constraints and production conditions not represented in this illustrative dataset.

Status

Illustrative proposal. No client decision recorded.

Responsibility statement

ScaleSight prepares the analysis and recommendations. Alec’s supplies context, makes the final decisions and executes through its existing systems.

Acceptance

The managed work is concrete.

- The workspace is not positioned as the product itself.
- Existing people and systems remain central.
- No implication that Alec’s is already a client.
- No dedicated full-time staffing claim.
- Pilot scope is specific.

08 Shared presentation, responsive & blocked states

Every recommendation must answer these six questions:

Field	Example
What deserves attention?	Collaboration production depth
What is proposed?	Compare 8,000 cups against the requested 12,000
Why?	Portfolio evidence, existing stock and competing capacity requirements
What makes it feasible?	Component and line-time checks
What could change it?	Stronger demand, different retailer timing or new commercial context
Who decides?	Alec’s team

Use separate confidence descriptions for:

• **Demand evidence**

• **Execution feasibility**

Use Low, Moderate or High with an explanation. Avoid invented confidence percentages.

Status:

Illustrative reviewed proposal - client decision not recorded.

Responsive and accessibility

Desktop

• Keep the main analysis and short interpretation close together.

• Use wide tables where comparison matters.

• Avoid oversized empty hero sections.

• Make product selection and active scenarios easy to see.

Mobile

• Lead with the business question and interpretation.

• Stack comparison panels.

• Provide intentional horizontal scrolling for detailed tables.

• Use a full-height evidence drawer.

• Keep disclosure and units readable.

• Collapse navigation into an accessible menu.

Accessibility

• Do not communicate status through color alone.

• Provide visible keyboard focus.

• Make tooltip content available by click and keyboard.

• Maintain readable contrast.

• Respect reduced-motion settings.

• Keep explanatory text available outside charts.

Tablet behavior

Comparative panels can stack. Product context remains visible. Controls remain usable without hover.

Number and chart rules

Measure	Display
Cups	`12,000 cups`
Money	`$18,600`
Line time	`36.83 hours`
Cover	`2.0 weeks`
Availability	`95.2%`
Forecast benchmark	`3.17%`

Every chart must identify:

• Unit.

• Time period.

• Retailer or inventory scope.

• Observed, forecast or simulated status.

Never combine retailer-owned stock with brand-owned stock in one total.

Never add firm orders on top of a forecast that already includes them.

Preserve calculation precision internally; round only for display.

Table rules

• Clear units in headers.

• Sticky headers where appropriate.

• Consistent SKU order.

• Product name alongside SKU code.

• Totals only for additive measures.

• No summed percentages or summed weeks of cover.

Missing and blocked states

Situation	Required response
Unsupported SKU-retailer combination	Explain the assortment assumption
Incomplete comparison	Show missing evidence
Supply beyond confirmed receipts	Mark future supply as uncommitted
Incompatible pack or destination	Block the option with a reason
Unknown payment terms	Show cost; mark cash timing unknown
Proposed analytical method	Label it as proposed
Zero denominator	Show “Not calculable”
Infeasible production	Show the binding constraints
Unreviewed exploration	Keep it separate from the reviewed proposal

Blocked states should help the user understand what information or permission would resolve the issue.

Exact explanatory messages

These are part of the demo’s credibility.

Unsupported retailer/product combination

This product is not eligible for the selected retailer in the illustrative assortment.

Insufficient comparison evidence

The comparison is incomplete. We would validate availability and commercial conditions before changing the commitment.

No new production confirmed beyond the horizon

Additional supply is not committed in this fixture. The projection identifies a next decision; it does not assume replenishment.

Incompatible transfer

The available stock has a different pack or destination restriction.

Missing payment terms

Payment timing is unresolved. The commitment value is shown separately.

Method not implemented

This method is proposed for a real-data investigation. The current demo uses the labeled synthetic comparison.

No denominator

Not calculated because the comparison denominator is zero.

No invented zeros should stand in for missing information.

Analyst copy and status vocabulary

Interpretation length

Use:

• Two or three sentences for the main review.

• One sentence for the client decision.

• Longer method explanation in the evidence drawer.

Do not bury the decision under a paragraph about ML.

Status vocabulary

Use consistently:

• Review production.

• Protect capacity.

• Hold additional depth.

• Reallocate.

• Resequence.

• Reserve component.

• Investigate.

• Watch.

Ownership labels

Suitable labels:

Prepared analytical comparison
Illustrative ScaleSight review
Decision for Alec’s
Requires client context

Avoid fake timestamps such as “analyst reviewed five minutes ago.”

Public context

Public research should appear where relevant:

• Collaboration identity on the launch investigation.

• In-house manufacturing context on production.

• Ingredient categories on components.

• Current retail focus on channel scope.

Do not paste a prospect research report onto the main workspace.

09 Build sequence & technical acceptance

Step 1 - Shared shell and data access

Build:

• Eleven routes.

• Navigation.

• Snapshot/disclosure.

• Shared product and scenario context.

• Evidence drawer.

• Calculation selectors.

Step 2 - Core decision journey

Build first:

1. Weekly Brief.

2. Incrementality.

3. Production.

4. Cash.

5. Managed Intelligence.

This is an implementation order, **not a reduction in scope**.

Step 3 - Supporting analytical depth

Build:

6. Reorder Quality.

7. Cohorts.

8. Components.

9. Allocation.

10. Forecast Learning.

Step 4 - Connected scenario behavior

Build:

11. Scenarios.

Then verify propagation across the other relevant pages.

Step 5 - Responsive and presentation QA

Check:

• Desktop.

• Tablet.

• Mobile.

• Direct route loading.

• Reset behavior.

• Product image mapping.

• Chart labels.

• Evidence links.

• Infeasible states.

• Numerical consistency.

Engine invariants

1. **One canonical dataset.** No page-specific copies of the same inventory or cost.
2. **Immutable reviewed baseline.** Exploratory scenarios are separate.
3. **Clear scopes.** Target launch panel, operating panel and control panel never silently merge.
4. **Separate ownership.** Retailer-owned inventory is not brand-owned available stock.
5. **Separate demand types.** POS, retailer receipts, orders and dispatch forecasts are distinct.
6. **No duplicate forecast/orders.** Firm orders consume forecast.
7. **No negative physical stock.** Track unmet demand separately.
8. **Respect timing.** Supply after a requirement cannot retroactively serve it.
9. **Respect physical compatibility.** SKU, pack, destination and lot conditions matter.
10. **Respect manufacturing constraints.** Hours, components and release dates must all work.
11. **Separate cost and cash.** Previously paid materials are not new payments.
12. **Do not turn estimates into facts.** Counterfactual movement is method-dependent.
13. **Do not turn simulation into review.** A changed scenario is not automatically specialist-approved.
14. **Explain infeasibility.** Never silently increase capacity to make an option work.
15. **Client retains judgment.** Recommendations remain proposed decisions.

Data-quality fixtures

Add a separate raw-input test feed. Do not corrupt the clean canonical tables.

Issue	Raw-input example	Required treatment
Duplicate retailer receipt	Same receipt imported twice	Deduplicate before inventory calculation
Unit mismatch	Cases entered as cups	Reject or convert using explicit unit metadata
Impossible availability	POS exists with zero available days	Quarantine for investigation
Unknown SKU	Unmapped source code	Exclude until mapped
Missing PO ETA	Receipt date absent	Do not assume arrival
Invalid transfer	CC01 destined for non-Target channel	Block
Pack incompatibility	Retail four-pack proposed for club demand	Block
Negative physical stock	Closing balance below zero	Separate unmet demand; flag reconciliation

Issue	Raw-input example	Required treatment
Forecast leakage	Forecast issued after actual week	Exclude from evaluation
Duplicate demand	Firm order added on top of full forecast	Consume forecast instead
Unpaid/paid confusion	Old deposit counted as future cash	Remove from future-payment total
Stale product metadata	Older description conflicts with rendered page	Prefer current source; preserve discrepancy note

These examples demonstrate that maintained analytical work includes validating inputs, not simply generating charts.

Numerical acceptance

These checks are required before release.

Demand and cohorts

```
Target prelaunch total = 10,000
Target latest total = 11,100
Latest incumbent total = 8,700
Latest collaboration = 2,400

Operating latest total = 13,600
Operating prelaunch total = 12,500

Cohort latest totals:
3,300 + 2,500 + 2,500 + 2,800 = 11,100

Counterfactual:
2,600 + 2,550 + 2,450 + 2,700 = 10,300

Associated difference:
11,100 - 10,300 = 800
```

Retail inventory

```
Collaboration retailer receipts = 14,880
Collaboration consumer POS = 10,900
Closing retailer stock = 3,980
```

Production

```
Requested output = 52,000
Reviewed output = 40,000
Bridge output = 40,000

Requested hours = 46.83
Reviewed hours = 36.83
Bridge hours = 37.33
```

Supply

```
42,000 + 40,000 + 40,000 - 105,200 = 16,800

No-reviewed-run unmet dispatch = 28,600
```

Components

```
Reviewed base requirement = 3,200 kg
Reviewed cup/lid requirement = 40,000
Reviewed granola = 96 kg

10,000 CC01 cups require 120 kg granola
12,000 CC01 cups require 144 kg granola
```

Allocation

```
East transferable surplus:
3,600 - 1,000 - 680 = 1,920

After transfer:
1,920 - 1,200 = 720

Network stock remains 6,000
```

Economics

```
Requested production cost = $56,900
Reviewed production cost = $43,700
Difference = $13,200

Reviewed conversion cash = $18,200
Transfer expense = $400
First-window future cash = $18,600

Existing PO future balances = $16,065
Bridge conversion cash = $17,600
Eight-week future cash = $52,265
```

Scenarios

```
Collaboration +25% unmet dispatch = 2,700
Cookies +20% unmet dispatch = 1,200
One-week reviewed-run delay = 6,300
```

Forecast evaluation

```
Actual launch-period portfolio = 45,800

Naive absolute error = 2,500
Reviewed illustrative absolute error = 1,450

Naive WAPE = 5.46%
Reviewed illustrative WAPE = 3.17%
```

This is the Phase 1 foundation: the eleven pages now share a coherent product, demand, inventory, production, component, allocation, finance and review model. Phase 2 can build the experience around it without inventing disconnected numbers.

10.A Source register

Public sources are retained from the supplied research snapshot of 11 October 2026. They were not freshly browsed for this document. Product metadata and dated founder statements require review before a real engagement.

S01 / Alec Jaffe / Purely Elizabeth collaboration founder post

https://www.linkedin.com/posts/alec-jaffe_you-officially-have-permission-to-eat-ice-activity-7510715727155347456-iT7t

S02 / Alec Jaffe / Culture Cup in-house production founder post

https://www.linkedin.com/posts/alec-jaffe_culturecup-alecsicecream-guthealth-activity-7315453670622433281-FGSg

S03 / Alec's Cookies & Cream product page / product FAQ

https://www.alecsicecream.com/products/culture-cup-cookies-cream-4-pack

S04 / Alec Jaffe / VP Finance founder post (role status unknown)

https://www.linkedin.com/posts/alec-jaffe_were-hiring-a-vp-of-finance-at-alecs-ice-activity-7466227760194056192-ebSz

S05 / Alec's $11M Series A announcement / 9 October 2025

https://www.prnewswire.com/news-releases/alecs-ice-cream-secures-11m-series-a-to-scale-nationwide-expansion-302578849.html

S06 / Alec's Culture Cup collection

https://www.alecsicecream.com/collections/culture-cup

S07 / Alec's x Purely Elizabeth collaboration page

https://www.alecsicecream.com/pages/alecs-x-purely-elizabeth

Source asset requirements

Use the eight supplied public product paths in the catalog to obtain genuine packaging assets during the build. Record asset URL, retrieval date, license/permission status and SKU mapping. Exact brand fonts and licensed assets are not supplied; do not label substitute typography or proposed tokens official.

10.B Assumptions & unresolved implementation points

ID	Assumption / boundary	Treatment
A01	Retailer panels, stores and historical launch date are synthetic	Keep sample scope visible; no real door-count claim
A02	Incumbent retailer permissions are fixture assumptions	CC01 public Target restriction is distinct
A03	Forecast, buffers and unchanged retailer inventory targets	Synthetic; firm orders consume forecast
A04	In-house run rate, changeovers, good-cup yield, run increments	Supplied planning assumptions; not factory evidence
A05	Base/component coefficients and all prices	Planning BOM only; not recipes
A06	Stock, lots and expiry	Synthetic; retain owner/QA/pack checks
A07	Bridge commitments, PO timing and supplier terms	Synthetic confirmed planning commitments only
A08	Previously paid materials and 30/70 PO split	Exclude paid amounts from future cash
A09	Transfer eligibility, transit and cost	Synthetic candidate; exact departure/arrival unspecified
A10	AP-106 cancellation/refund terms	Unknown; do not assume reversibility
A11	Forecast issue dates and promo/price records	Add technical chronology; missing commercial data stays null
A12	Counterfactual comparability	Assumption sensitivity; no causal or confidence-interval claim
A13	Club pack and order	Hypothetical; blocked until separately defined
A14	Proposed colors and substitute font	Not certified official Alec's brand tokens
A15	Store allocation and historical control interpolation	Added deterministic developer rules in Appendix C
A16	Lot/node partition outside the CC04 worked case	Unspecified; aggregate scope until client/fixture rule supplied
A17	Review identity, approvals and booking destination	No invented client approval or fake booking

Transfer date versus cash date

The source dates the transfer expense on 26 October but requires stock to arrive before West's projected depletion during the first two weeks. Treat 26 October as the payment date, not the physical arrival date. Exact dispatch/arrival dates are not supplied. Enforce the two-day transit plus one-day release constraint and show unresolved timing until a date is explicitly selected. Do not present the payment date as proof of service feasibility.

Pack granularity and physical movements

Consumer POS is integer four-packs, and the supplied receipt events/transfer are 24-cup case compatible. Forecast expected cups may be fractional. Several aggregate run, stock and dispatch anchors are cup-equivalent planning quantities rather than full logistics-case quantities. Do not silently round canonical anchors to whole cases. During implementation, distinguish analytical requirements from executable case-rounded shipments and explicitly expose any residual if case-only execution is imposed.

Optional reservation and requested plan

AP-106 is uncommitted and is excluded from the baseline payment schedule. Requested cash is at least $29,965 before expedited freight, overtime, extra capacity and any transfer expense; it is not a feasible like-for-like executed alternative. Source prices and order multiples do not establish that top-ups can arrive before the first window.

Underdetermined node simulation

The source defines the CC04 regional worked case but not a full SKU-by-node dispatch and receipt matrix. Implement the supplied eight-week inventory projection at aggregate SKU scope. For a full node-level engine, retain unknown allocations as null or adopt a separately labeled synthetic node-partition rule. Do not represent an unspecified node allocation as verified or client-approved.

10.C Deterministic store-panel generation specification

The dense panel is an implementation fixture specification. It has not been generated in this document workflow. The rules below supplement the source's seed 241 and largest-remainder allocation with explicit technical tie-breaks. They are developer assumptions, never company data.

Dimensions and eligibility

280 stores x 8 SKUs x 20 weeks = 44,800 rows. Operating IDs: T001-T120, WF001-WF060 and WG001-WG040. Control IDs: TC001-TC060. Generate all rows, including zero nonlisted combinations. CC01 is unlisted before 14 September, at non-Target retailers and in the control panel. CC02-CC08 are listed under the synthetic assortment. Operating/control membership is immutable.

Fixed weights and rounding

Use a local pseudorandom integer generator with seed 241; persist its algorithm/version. Assign each sorted store a fixed positive integer weight. Use largest-remainder rounding in four-packs; ties resolve by ascending stable ID. Weights stay fixed across all weeks and SKUs. Record generationVersion and the additional rule IDs.

```
# Added developer choice: portable deterministic LCG, positive weights.
state = 241
for store in sorted(store_ids):
    state = (1664525 * state + 1013904223) % (2 ** 32)
    weight[store] = 50 + state % 101

# Allocate integer totals without float-rounding drift.
def allocate(total_packs, ids, weights):
    denominator = sum(weights[i] for i in ids)
    q = {i: total_packs * weights[i] // denominator for i in ids}
    remainder = {i: total_packs * weights[i] % denominator for i in ids}
    residual = total_packs - sum(q.values())
    ranked = sorted(ids, key=lambda i: (-remainder[i], i))
    for i in ranked[:residual]: q[i] += 1
    return q
```

Preserve every canonical anchor

1. For the first twelve weeks, allocate Target total packs across CC02-CC08 using the supplied 10,000-cup prelaunch mix and the exact total multipliers. For 17 August-5 October use every explicit SKU total.

2. For each incumbent SKU, use prelaunch cohort weights before launch and latest cohort weights during launch, with largest-remainder pack allocation. Preserve the exact prelaunch and latest cohort matrices. For CC01 use all four explicit launch-week cohort totals.

3. Within each Target cohort allocate SKU packs across its 30 stores using fixed weights. Whole Foods/Wegmans use their exact constant retailer-SKU quantities every historical week, allocated over their store IDs.

4. Comparison groups have 15 stores each, and CC01 remains zero. Use 1,000 cups/group in prelaunch weeks and the supplied latest control totals of 1,040/1,020/980/1,080. Added interpolation rule: intermediate launch totals linearly interpolate from 1,000 to the latest total in integer four-packs, then allocate incumbents by the prelaunch SKU shares. Earlier control weeks remain 1,000/group. This maintains the latest counterfactual anchors without claiming supplied historical controls.

5. Latest CC01 available days must sum to 200/196/168/204 for A/B/C/D. For a deterministic technical distribution, assign floor(totalDays/30) per store plus one to the first remainder stores by ID. For other listed rows use seven available days as a labeled controlled fixture assumption. For nonlisted rows use zero. Match

control-panel exposure proportionately at group level for the supplied counterfactual; do not claim exact customer comparability.

6. Keep promoFlag and priceIndex null where commercial context is not supplied. Receipt events remain a separate cohort-level table; do not invent per-store receipts. Mark the fact that latest availability normalization is supported, while broader commercial-event context is incomplete.

Required generation validation

Require exactly 44,800 unique store-SKU-week keys; integer nonnegative packs; cups = packs x 4; no nonlisted sales; positive available days for rows with POS; exact weekly retailer-SKU totals; exact cohort anchors and available-day sums; exact four-week collaboration totals; exact control-panel latest totals. Exclude all 9,600 control rows from operating demand. Keep the raw quality-test feed separate.

Representative record contract

```
{
  "snapshotId": "ALECS-DEMO-2026-10-12-V1",
  "storeId": "T001", "skuId": "CC01", "week": "2026-10-05",
  "packsSold": "integer produced by allocate()",
  "cupsSold": "packsSold * 4", "listed": true,
  "availableDays": 7, "promoFlag": null, "priceIndex": null,
  "evidenceClass": "SYNTHETIC INPUT", "operatingOrControl": "operating"
}
```

This is a schema example, not a claimed generated numeric store record. A nonlisted WF/CC01 row has packsSold=0, cupsSold=0, listed=false and availableDays=0. Store T001 is in the first 20 A stores with seven days under the added availability rule.

10.D Formula and selector reference

Selector / calculation	Formula / contract	Scope
observedPortfolioGrowth	11,100 - 10,000 = 1,100	Latest Target panel vs prelaunch
incumbentMovement	8,700 - 10,000 = -1,300	Excludes CC01
netGrowthRatio	1,100 / 2,400 = 45.8%	Descriptive, not causal
availableVelocity	POS cups / (available days / 7)	Cohort-SKU-week
counterfactual	prelaunch cohort x latest control / baseline control	Matched synthetic groups
associatedDifference	11,100 - 10,300 = 800	Assumption-dependent estimate
retailClosing	opening + receipts - POS - writeoffs	Retailer ownership only
requiredReceipts	consumer demand + target ending retail stock - opening retail stock	Consume firm orders from forecast
stockAvailable	opening + released supply + inbound - outbound - writeoffs	Eligible brand-owned stock
served / unmet	min(required, available) / required - served	Period; no automatic backlog
physicalClosing	available - served	Nonnegative
componentRequirement	sum(output x coefficient)	Evaluated option only
productionHours	cups/1200 + max(activeRuns-1,0) x 0.5	Reserved window
transferableSurplus	eligible stock - protected demand - donor buffer	Eligible donor node
productionCost	material consumption + conversion	USD, not future cash
futureCash	unpaid procurement + conversion + incremental logistics	Dated obligations
WAPE	sum(abs(forecast-actual)) / sum(actual)	Null for zero actual total
signedBias	sum(forecast-actual) / sum(actual)	Positive = overforecast

Forward buffer implementation

Added technical rule: buffer cups equal the next policyWeeks of forecast requirement, summing complete weeks plus a fraction of the next week for the 1.5-week CC01 policy. Use extended W9-W13 demand when available. Where a required future period is outside the supplied horizon, return incomplete rather than zero. Buffer breach is separate from unmet dispatch.

Forecast chronology

Added technical rule for the synthetic archived ledger: issue each forecast before its target week's start and observe actuals after the week's end. The exact issue timestamp is not supplied; use a documented synthetic prior-week date if a concrete timestamp is required. Reject leakage. Preserve all supplied forecast values; do not fit a model and silently replace the ledger.

Source tables and duplication policy

All complete canonical numerical tables are printed in Phase 1. Each per-page brief and UX chapter refers to that same snapshot. The appendix serves as an index and formula reference; it introduces no competing copy of inventory or cost tables. Display precision must never alter canonical quantities.